

Graph Theory

Rosen 8th ed., ch. 10

欧拉简介

- 莱昂哈德·欧拉（1707年~1783年），生于瑞士巴塞尔（**Basel**）
- 数学史上公认的四名最伟大的数学家之一；阿基米德、牛顿、欧拉和高斯
- 他是一位数学家、还在力学、光学、天文学等领域有突出贡献。
- 欧拉十五岁大学毕业；十六岁已经硕士毕业，十九岁博士毕业，毕业论文是一篇物理论文。二十岁欧拉被著名学家约翰·伯努利（雅各布·伯努利—概率论先驱之弟）的儿子丹尼尔·伯努利邀请到俄国，任生理学院院长
- 他一生写下886种书籍论文
- 59岁时因病致盲，此后口述完成了400多篇论文和多部论著

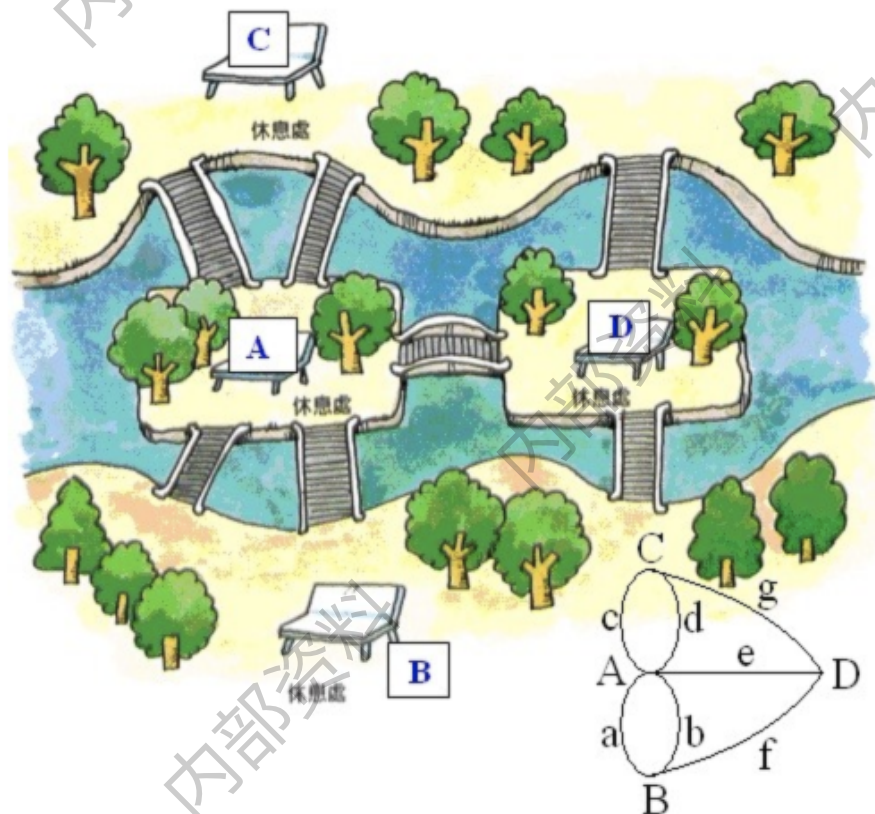




Königsberg Bridge problem

柯尼斯堡七桥问题

- 柯尼斯堡 (**Königsberg**) : 哥尼斯堡是东普鲁士的首都, 今俄罗斯加里宁格勒市
- 1736年, 发表论文给出解答
- 图论之父



10.1 图的概念/Introduction of Graph

10.2 图的术语/Graph Terminology

10.3 图的表示与同构/

Representing Graph and Graph Isomorphism

10.4 连通性/Connectivity

10.5 欧拉道路与哈密尔顿道路/

Euler and Hamilton Paths

10.6 最短道路问题/Shortest Path Problem

10.7 平面图/Planar Graphs

10.8 图的着色/Graph Coloring

Transport networks传输网流量问题

What are Graphs?

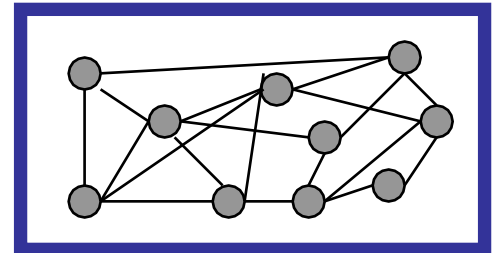
- General meaning in everyday math:
A plot or chart of numerical data using a coordinate system.



Not Our
Meaning

Technical meaning in discrete mathematics:

A particular class of discrete structures (to be defined) that is useful for representing relations and has a convenient webby-looking graphical representation.



Definition:

A graph $G = (V, E)$ consists of V , a nonempty set of vertices (or nodes) and E , a set of edges.

Each edge has either one or two vertices associated with it, called its endpoints. An edge is said to connect its endpoints.

Remark: The set of vertices V of a graph G may be infinite. A graph with an infinite vertex set or an infinite number of edges is called an **infinite graph**, and in comparison, a graph with a finite vertex set and a finite edge set is called a finite graph. In this book we will usually consider only **finite graphs**.

Applications of Graphs

- Potentially anything (graphs can represent relations, relations can describe the extension of any predicate).
- Apps in networking, scheduling, flow optimization, circuit design, path planning.
- More apps: Geneology analysis, computer game-playing, program compilation, object-oriented design, ...

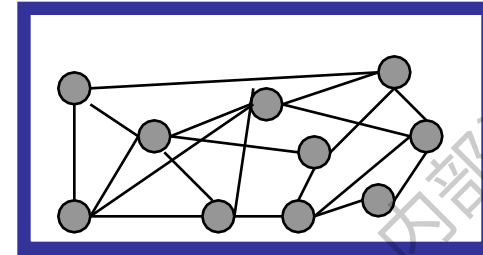
Types of Graphs

- Simple graph (简单图)
 - Multigraph (多重图)
 - Pseudograph (伪图)
 - Directed graph (有向图)
 - Directed multigraph (有向多重图)
- Undirected
- Directed

Simple Graphs

简单图

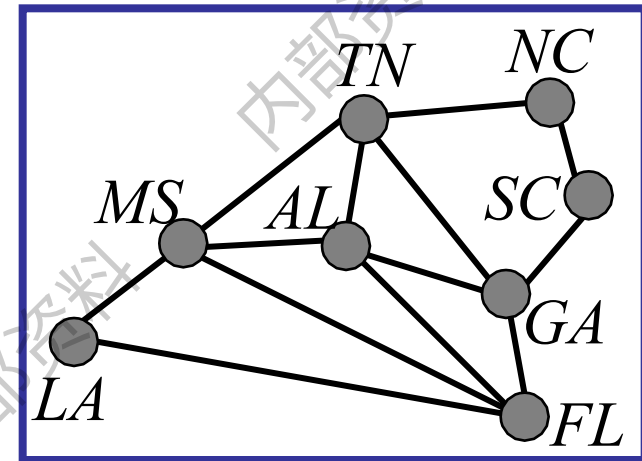
- Correspond to symmetric, irreflexive binary relations R .
- A simple graph $G=(V,E)$ consists of:
 - V , a nonempty set of *vertices* or *nodes*
 - E , a set of *edges* / *arcs* / *links*: unordered pairs of distinct elements $u,v \in V$, such that uRv .



*Visual Representation
of a Simple Graph*

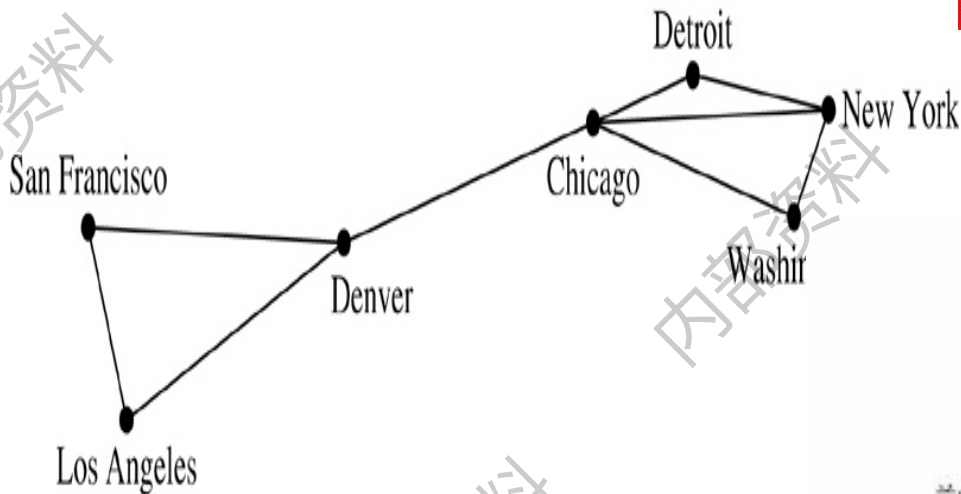
Example of a *Simple Graph*

- Let V be the set of states in the far-southeastern U.S.:
 - /e., $V = \{FL, GA, AL, MS, LA, SC, TN, NC\}$
- Let $E = \{\{u, v\} | u \text{ adjoins } v\}$
 $= \{\{FL, GA\}, \{FL, AL\}, \{FL, MS\},$
 $\{FL, LA\}, \{GA, AL\}, \{AL, MS\},$
 $\{MS, LA\}, \{GA, SC\}, \{GA, TN\},$
 $\{SC, NC\}, \{NC, TN\}, \{MS, TN\},$
 $\{MS, AL\}\}$



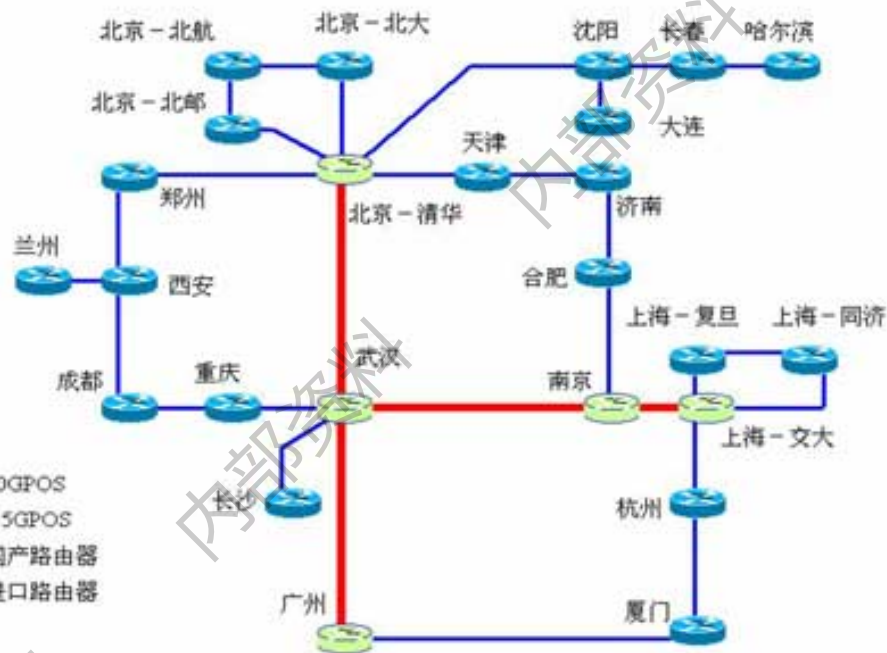
Computer Network

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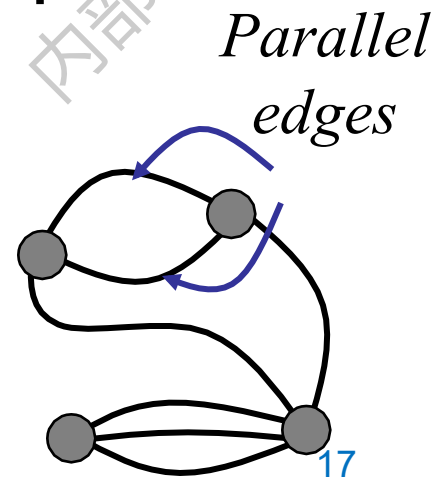
CERNET



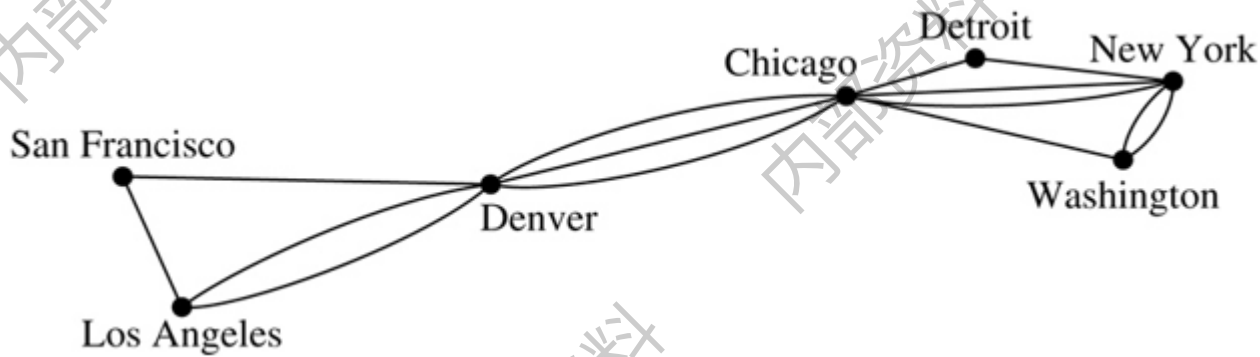
Multigraphs

多重图

- Like simple graphs, but there may be *more than one* edge connecting two given nodes.
- A *multigraph* $G=(V, E, f)$ consists of a set V of vertices, a set E of edges (as primitive objects), and a function $f:E \rightarrow \{\{u,v\} | u,v \in V \wedge u \neq v\}$.
- *E.g.*, nodes are cities, edges are segments of major highways.



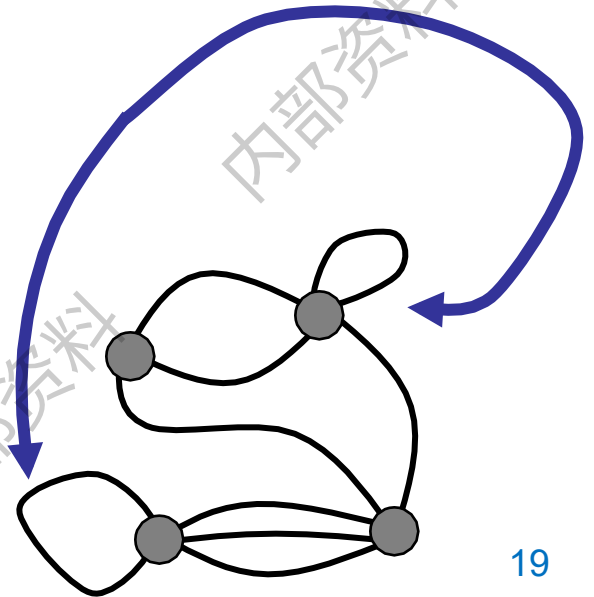
Highway Network with Multiple Links



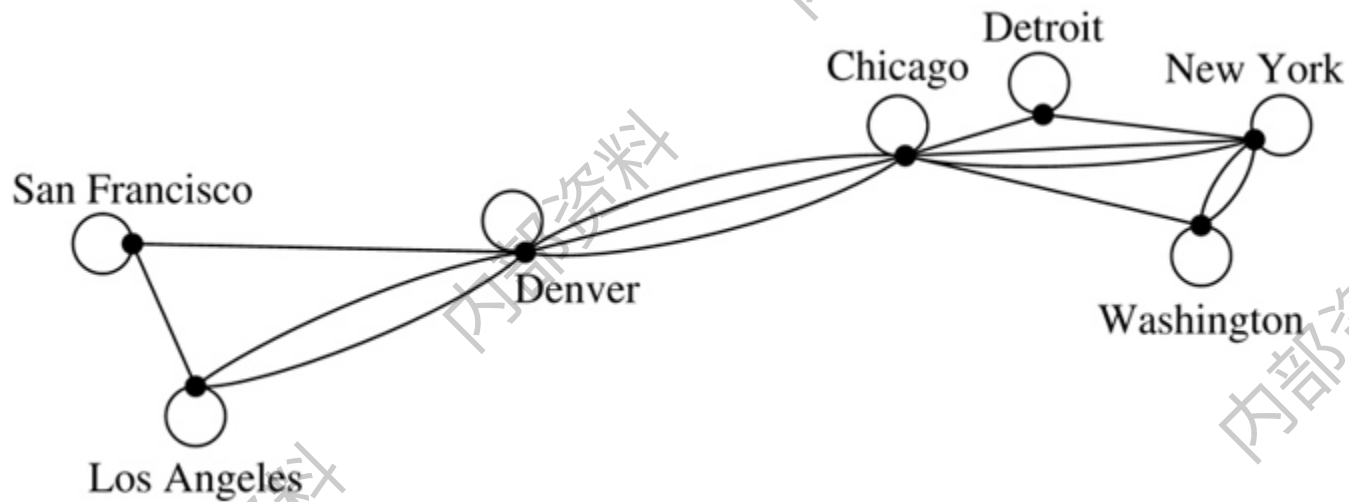
Pseudographs

伪图

- Like a multigraph, but edges connecting a node to itself are allowed.
- A pseudograph $G=(V, E, f)$ where $f:E \rightarrow \{\{u,v\} | u,v \in V\}$. Edge $e \in E$ is a *loop* if $f(e)=\{u,u\}=\{u\}$.
- E.g., nodes are campsites in a state park, edges are hiking trails through the woods.



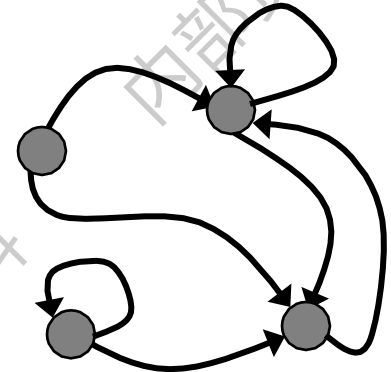
Computer Network with Diagnostic Links



Directed Graphs

有向图

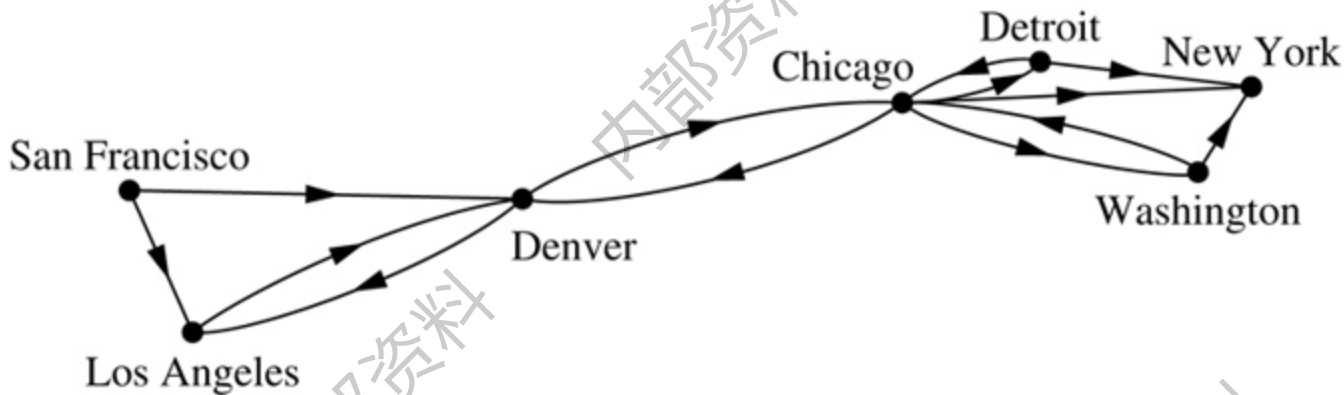
- Correspond to arbitrary binary relations R , which need not be symmetric.
- A *directed graph* (V, E) consists of a set of vertices V and a binary relation E on V .
- E.g.: $V =$ set of People,
 $E = \{(x, y) \mid x \text{ loves } y\}$



Simple directed graph

A directed graph has no loops and has no multiple directed edges.

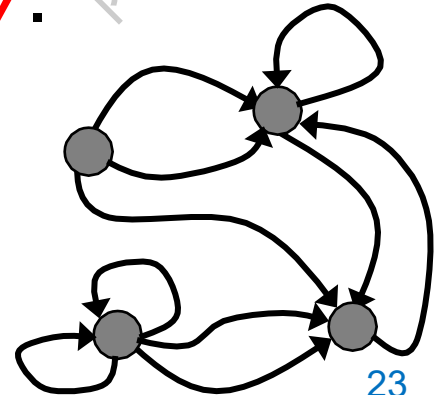
Television Network with One-way Links



Directed Multigraphs

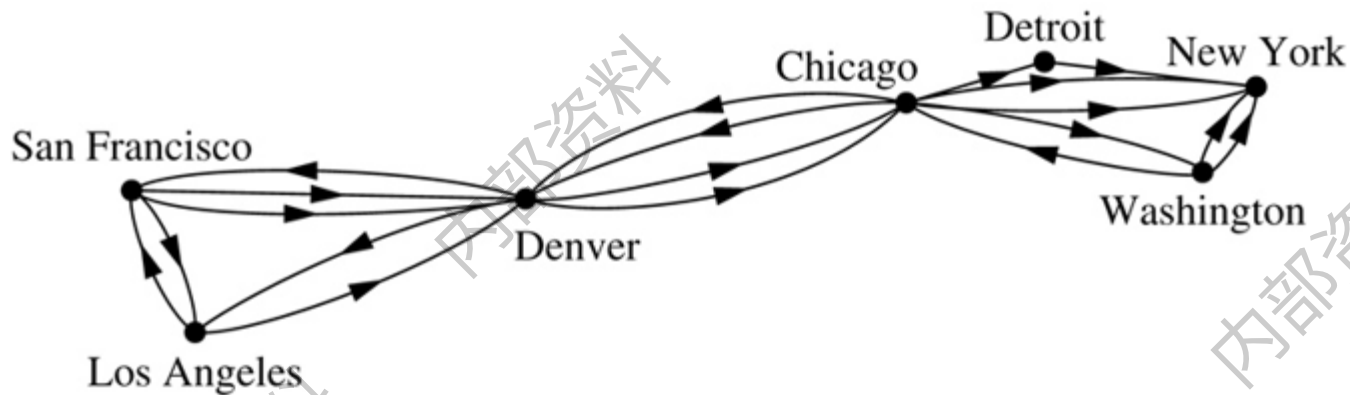
有向多重图

- Like directed graphs, but there may be more than one arc from a node to another.
- A *directed multigraph* $G=(V, E, f)$ consists of a set V of vertices, a set E of edges, and a function $f:E \rightarrow V \times V$.
- *E.g.*, V =web pages, E =hyperlinks. The WWW is a directed multigraph...





Television Network with Multiple One-way Links



Mixed Graphs

混合图

- A graph with both directed edges and undirected edges is called mixed graph

Types of Graphs: Summary

图类型

- Summary of the book's definitions.
- Keep in mind this terminology is not fully standardized across different authors...

TABLE 1 Graph Terminology.

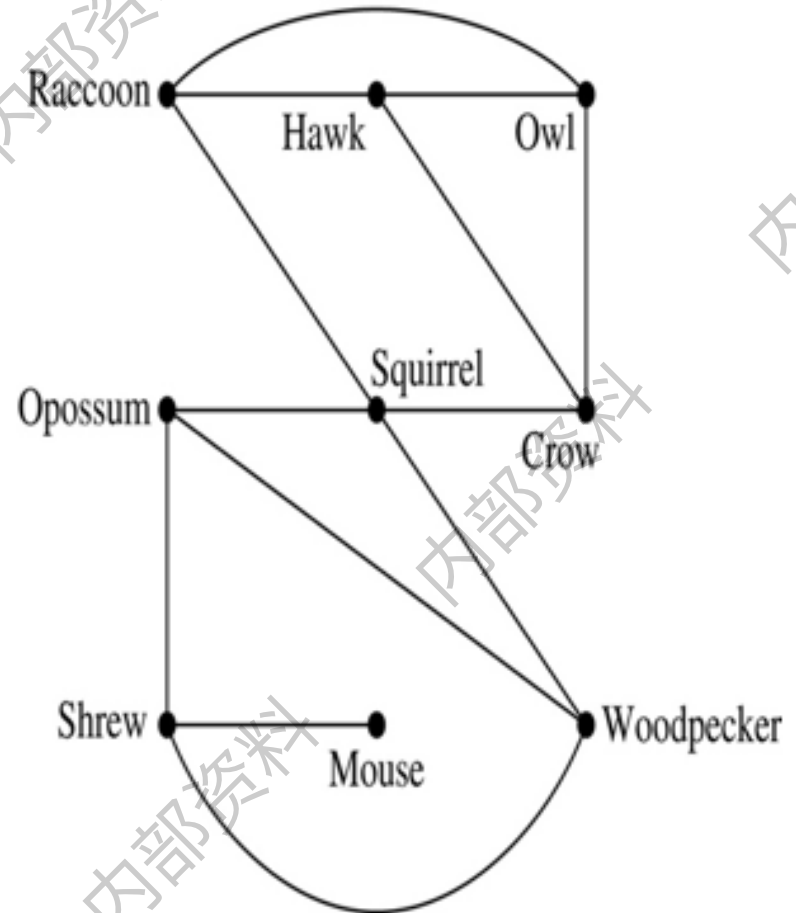
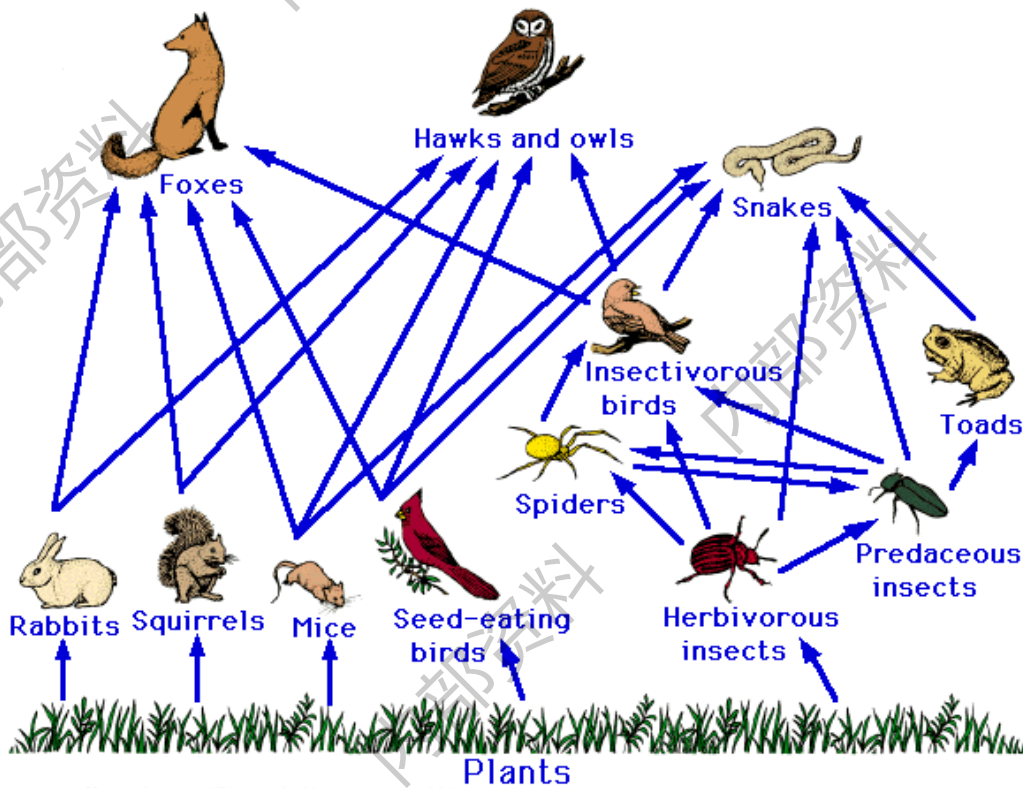
<i>Type</i>	<i>Edges</i>	<i>Multiple Edges Allowed?</i>	<i>Loops Allowed?</i>
Simple graph	Undirected	No	No
Multigraph	Undirected	Yes	No
Pseudograph	Undirected	Yes	Yes
Simple directed graph	Directed	No	No
Directed multigraph	Directed	Yes	Yes
Mixed graph	Directed and undirected	Yes	Yes

Graph Models

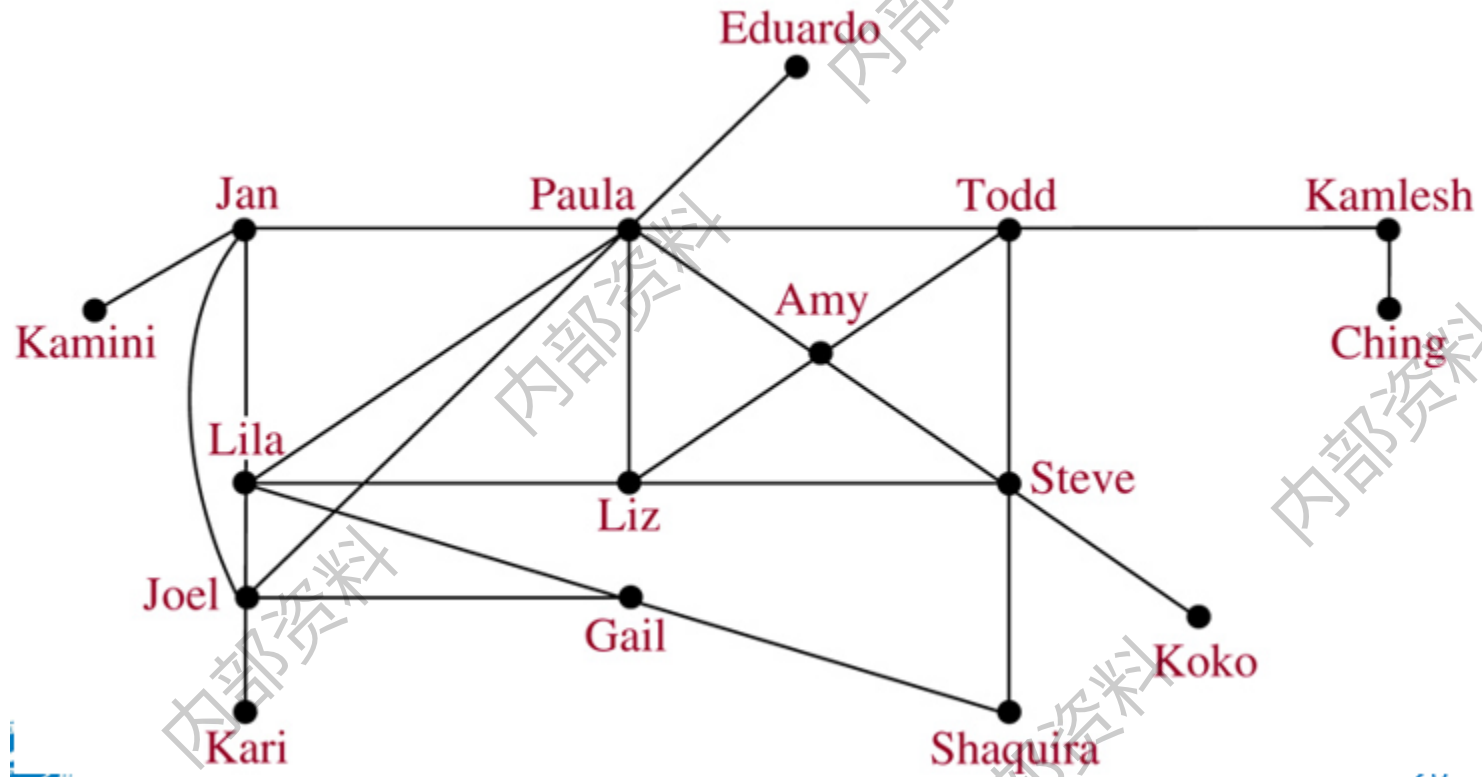
图模型

Niche overlap graphs in Ecology

生态图

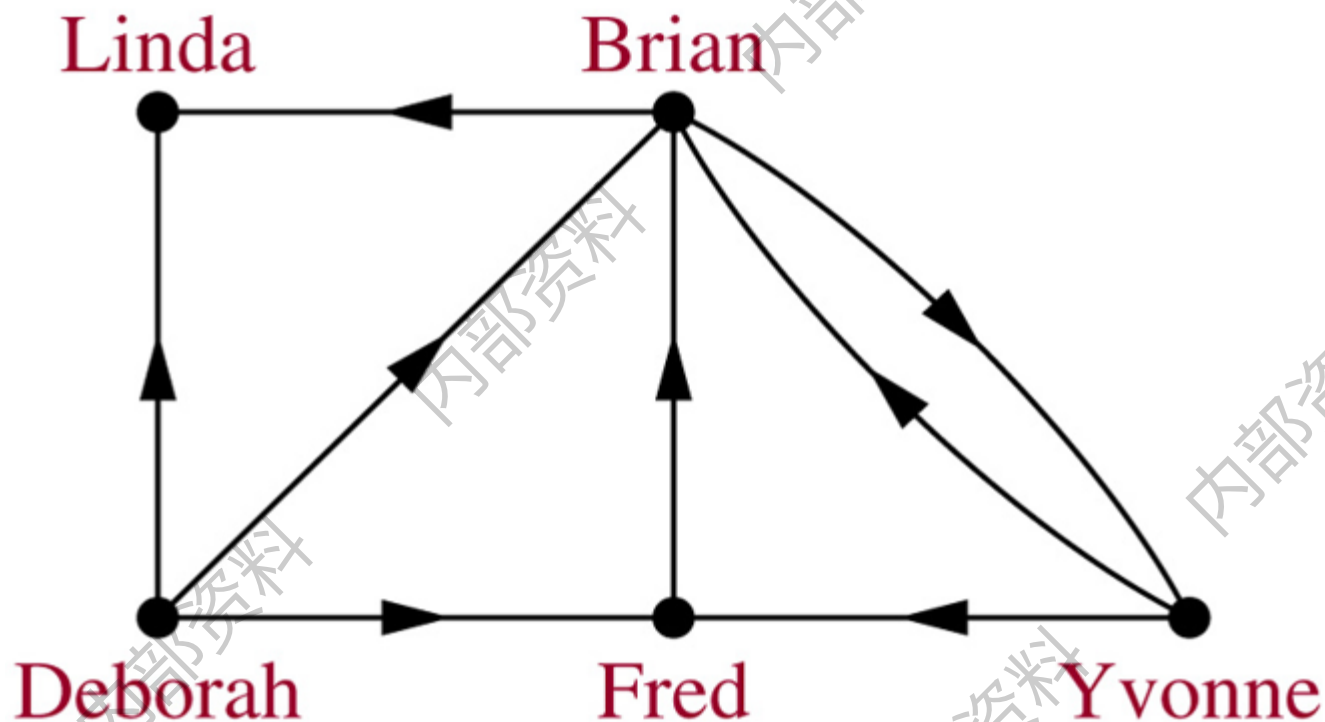


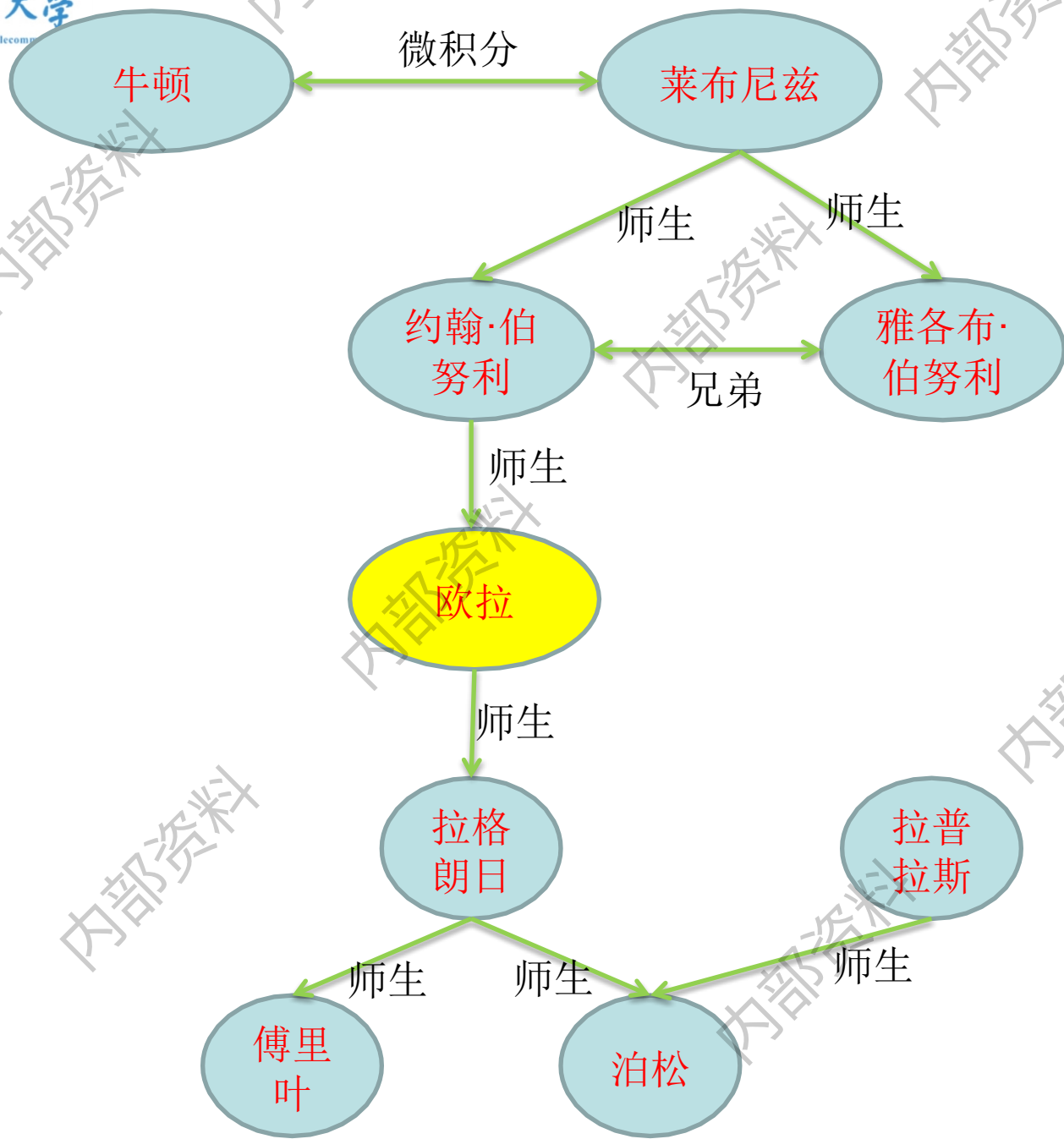
Acquaintanceship graphs 相识图



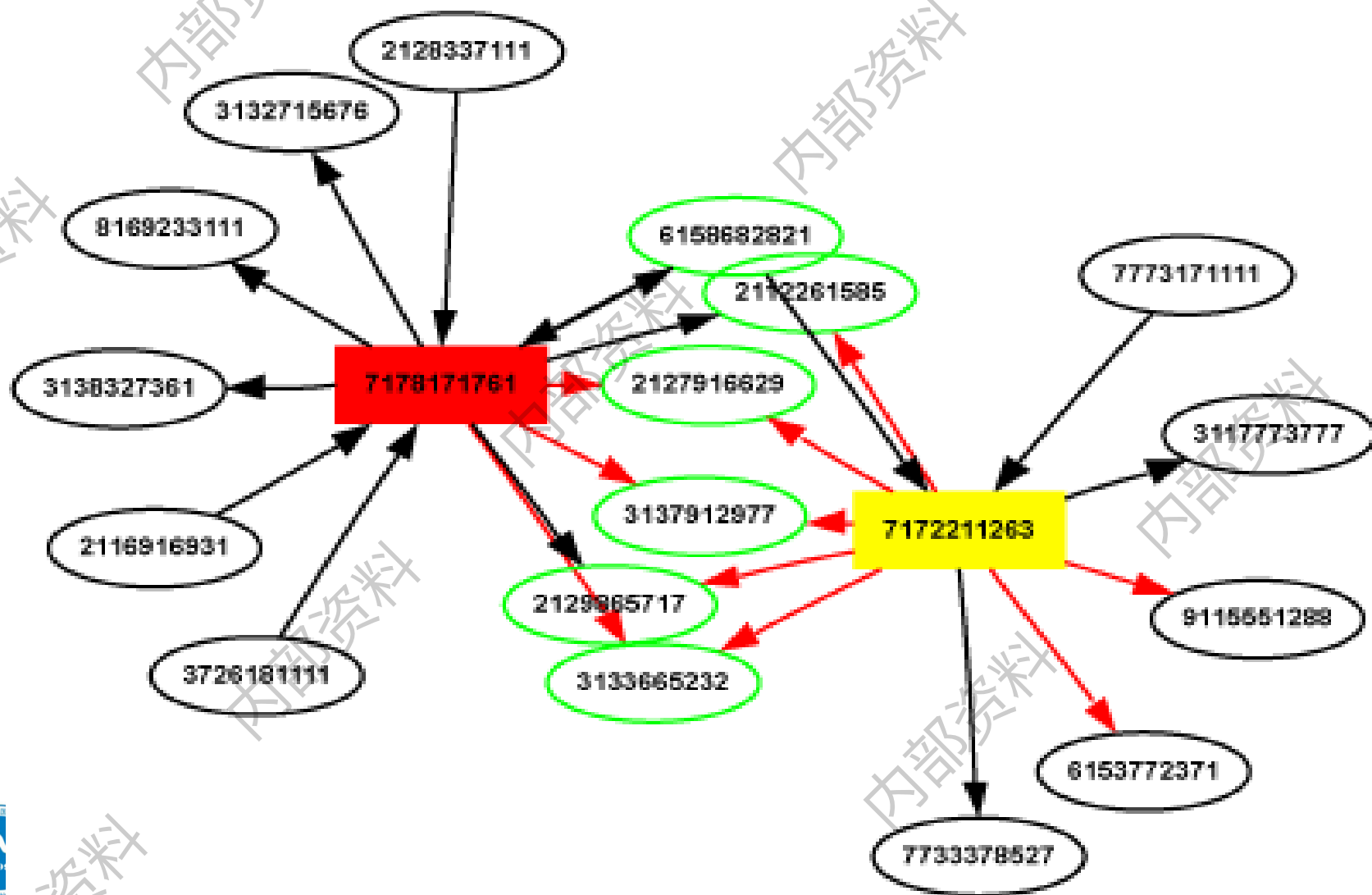
Influence Graph

影响力图

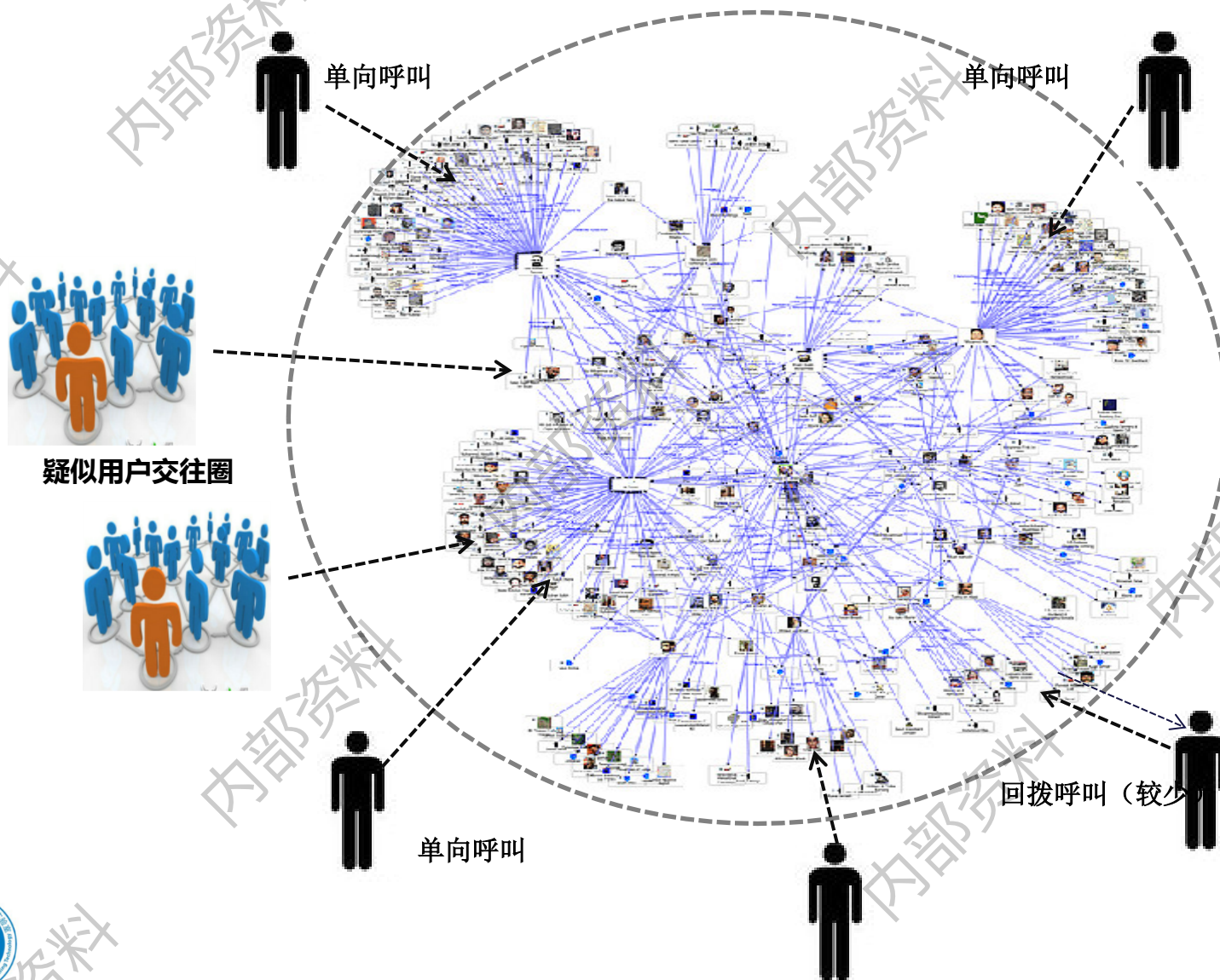




Call Graphs

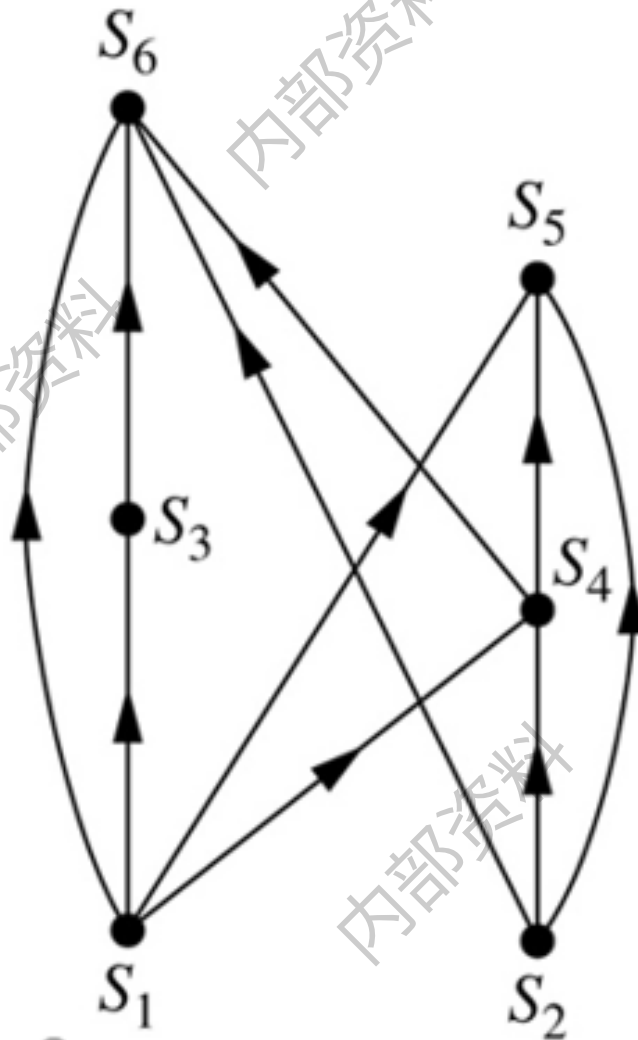


电信诈骗关系图

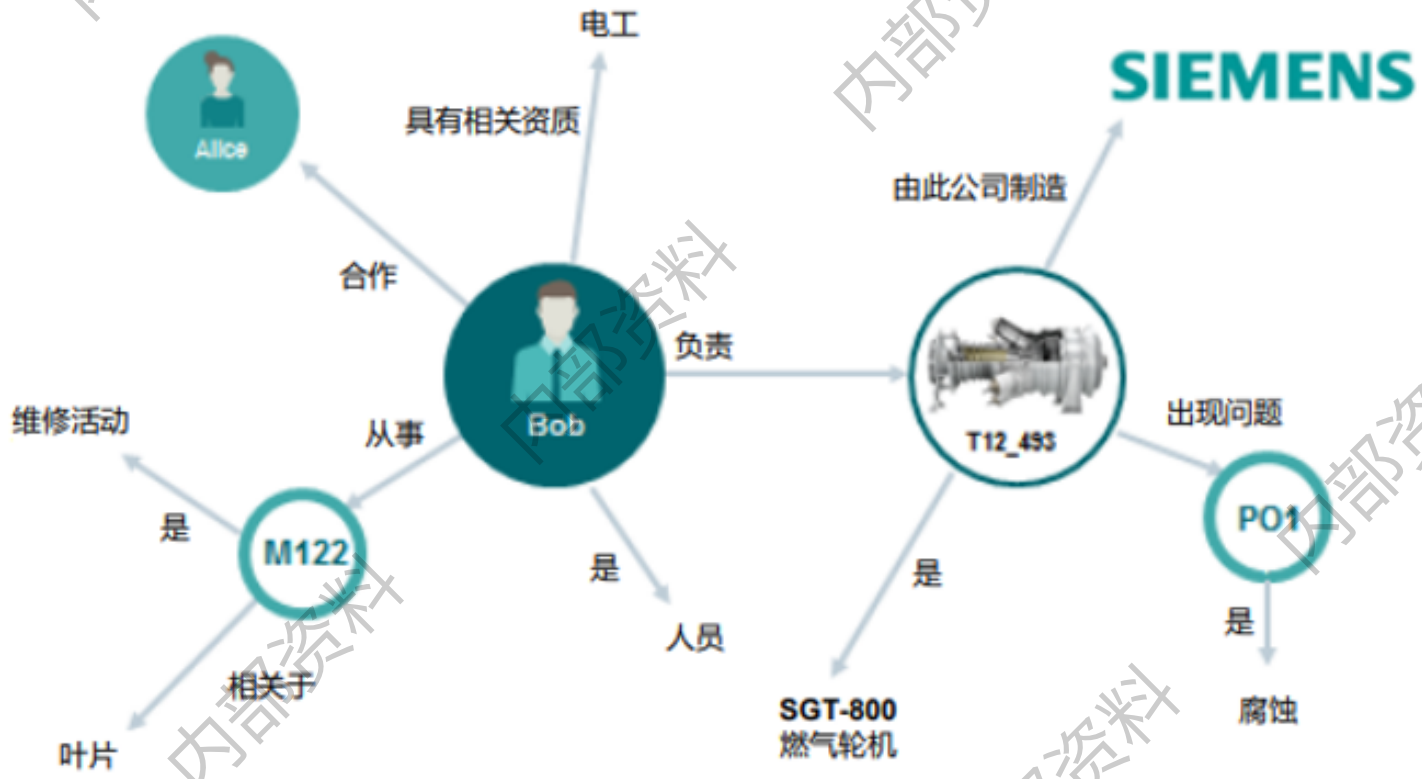


Precedence graph of statement execution 并发处理中的优先性图

S_1 $a := 0$
 S_2 $b := 1$
 S_3 $c := a + 1$
 S_4 $d := b + a$
 S_5 $e := d + 1$
 S_6 $e := c + d$



知识图谱 (Knowledge Graph)



§10.2: Graph Terminology

You need to learn the following terms:

- *Adjacent, connects, endpoints, degree, initial, terminal, in-degree, out-degree, complete, cycles, wheels, n-cubes, bipartite, subgraph, union.*

Adjacency (邻接)

Let G be an undirected graph with edge set E . Let $e \in E$ be (or map to) the pair $\{u, v\}$.

Then we say:

- u, v are *adjacent / neighbors / connected*.
- Edge e is *incident with* vertices u and v .
- Edge e *connects* u and v .
- Vertices u and v are *endpoints* of edge e .

Neighborhood

邻域

- The set of all neighbors of a vertex v of $G = (V, E)$, denoted by $N(v)$, is called the *neighborhood* of v .
- If A is a subset of V , we denote by $N(A)$ the set of all vertices in G that are adjacent to at least one vertex in A . So, $N(A) = \bigcup_{v \in A} N(v)$.

Degree of a Vertex

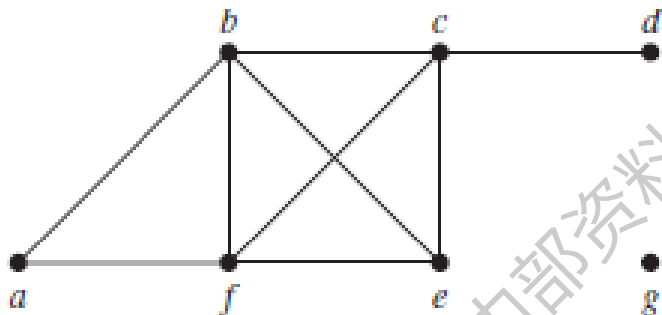
顶点的度

- Let G be an **undirected** graph, $v \in V$ a vertex.
- The *degree* of v , **$\deg(v)$** , is its number of incident edges. (Except that any self-loops are counted twice.)
- A vertex with degree 0 is called *isolated*.
(孤立点)
- A vertex of degree 1 is called *pendant*.
(悬挂点)

Example 1

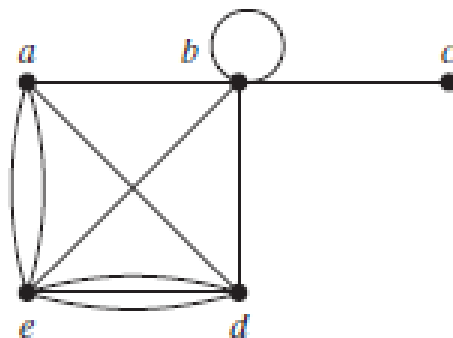
$$\deg(a) = 2, \deg(b) = \deg(c) = \deg(f) = 4,$$

$$N(e) = \{b, c, f\}, N(f) = \{a, b, c, e\}$$



G

$$\deg(a) = 4, \deg(b) = \deg(e) = 6$$



H

FIGURE 1 The Undirected Graphs *G* and *H*.

Handshaking Theorem

握手定理

- Let G be an undirected (simple, multi-, or pseudo-) graph with vertex set V and edge set E . Then

$$\sum_{v \in V} \deg(v) = 2|E|$$

- Corollary:** Any undirected graph has an even number of vertices of odd degree.

度序列可简单图化？

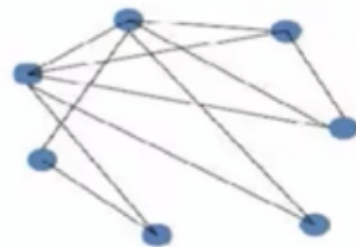
例：度序列 $(5, 3, 3, 2, 1)$
是否可图化？
是否可简单图化？



可简单图化的充要条件

- **Havel定理:** 设非负整数数列 $d=(d_1, d_2, d_3, \dots, d_n)$, 满足 $d_1 + d_2 + \dots + d_n = 0 \pmod{2}$, $n-1 \geq d_1 \geq d_2 \geq d_3 \geq \dots \geq d_n \geq 0$, 则 d 可简单图化 $\Leftrightarrow d' = (d_2 - 1, d_3 - 1, \dots, d_n - 1)$ 可简单图化

例: $(5, 5, 3, 3, 2, 2, 2), (4, 2, 2, 1, 1, 2), (4, 2, 2, 2, 1, 1),$
 $(1, 1, 1, 0, 1), (1, 1, 1, 1), (0, 1, 1), (1, 1)$



Directed Adjacency

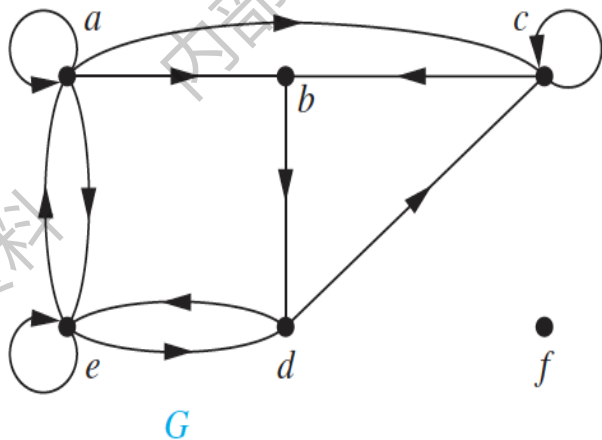
有向邻接

- Let G be a directed (possibly multi-) graph, and let e be an edge of G that is (or maps to) (u, v) . Then we say:
 - u is *adjacent to* v , v is *adjacent from* u
 - e comes from u , e goes to v .
 - e connects u to v , e goes from u to v
 - the *initial vertex* of e is u
 - the *terminal vertex* of e is v

Directed Degree

- Let G be a directed graph, v a vertex of G .
 - The *in-degree* of v , $\deg^-(v)$, is the number of edges going to v . 入度
 - The *out-degree* of v , $\deg^+(v)$, is the number of edges coming from v . 出度
 - The *degree* of v , $\deg(v) \equiv \deg^-(v) + \deg^+(v)$, is the sum of v 's in-degree and out-degree.

Example 4



$$\deg^-(a) = 2, \quad \deg^+(a) = 4,$$

$$\deg^-(b) = 2, \quad \deg^+(b) = 1,$$

$$\deg^-(f) = 0, \quad \deg^+(f) = 0.$$

FIGURE 2 The Directed Graph G .

Solution: The in-degrees in G are $\deg^-(a) = 2$, $\deg^-(b) = 2$, $\deg^-(c) = 3$, $\deg^-(d) = 2$, $\deg^-(e) = 3$, and $\deg^-(f) = 0$. The out-degrees are $\deg^+(a) = 4$, $\deg^+(b) = 1$, $\deg^+(c) = 2$, $\deg^+(d) = 2$, $\deg^+(e) = 3$, and $\deg^+(f) = 0$.

Directed Handshaking Theorem

有向图的握手定理

- Let G be a directed (possibly multi-) graph with vertex set V and edge set E . Then:

$$\sum_{v \in V} \deg^{-}(v) = \sum_{v \in V} \deg^{+}(v) = \frac{1}{2} \sum_{v \in V} \deg(v) = |E|$$

- Note that the degree of a node is unchanged by whether we consider its edges to be directed or undirected.

Underlying undirected graph

基本无向图

- The undirected graph that results from ignoring directions of edges is called the underlying undirected graph.

Special Graph Structures

特殊图结构

Special cases of undirected graph structures:

- Complete graphs (完全图) K_n
- Cycles (环图) C_n
- Wheels (轮图) W_n
- n -Cubes (n 立方图) Q_n
- Bipartite graphs (二部图)
- Complete bipartite graphs (完全二部图) $K_{m,n}$

Complete Graphs

完全图

- For any $n \in \mathbf{N}$, a *complete graph* on n vertices, K_n , is a simple graph with n nodes in which every node is adjacent to every other node: $\forall u, v \in V: u \neq v \leftrightarrow \{u, v\} \in E$.

K_1

K_2

K_3

K_4

K_5

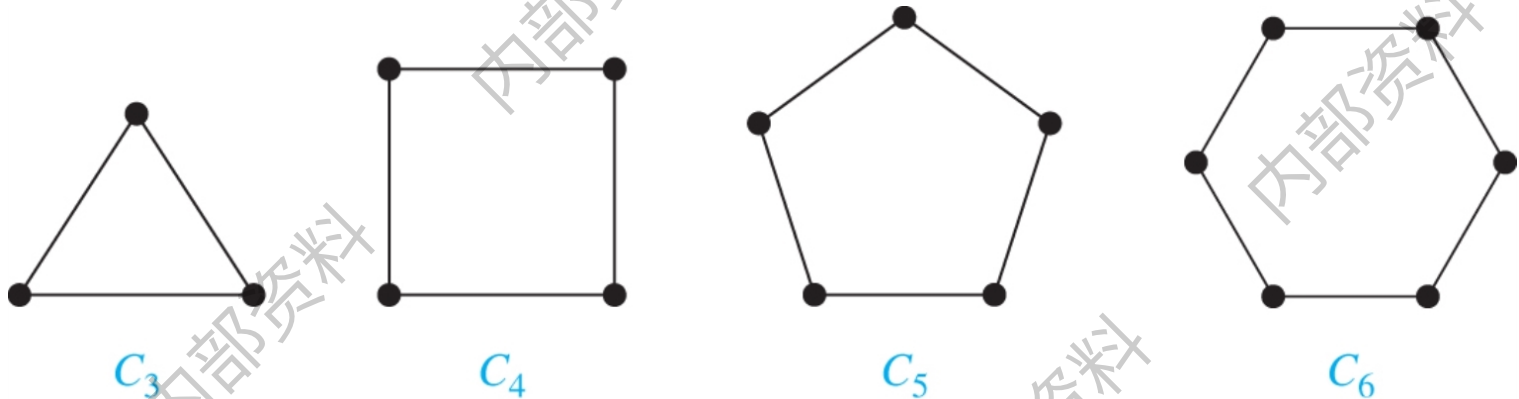
K_6

Note that K_n has $\sum_{i=1}^{n-1} i = \frac{n(n-1)}{2}$ edges.

Cycles

环图

- For any $n \geq 3$, a cycle on n vertices, C_n , is a simple graph where $V = \{v_1, v_2, \dots, v_n\}$ and $E = \{\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}, \{v_n, v_1\}\}$.

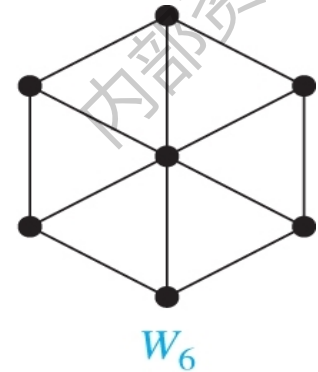
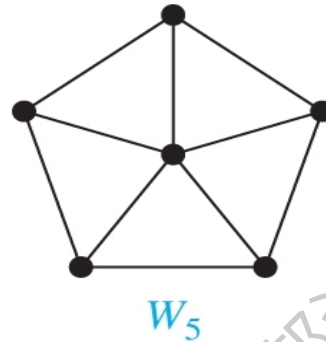
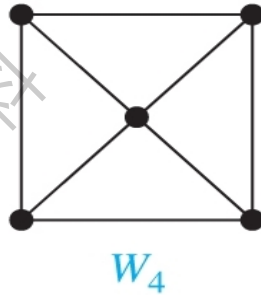
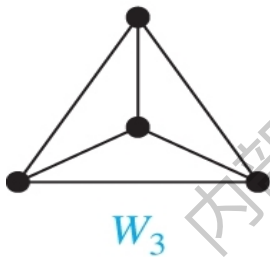


How many edges are there in C_n ?

Wheels

轮图

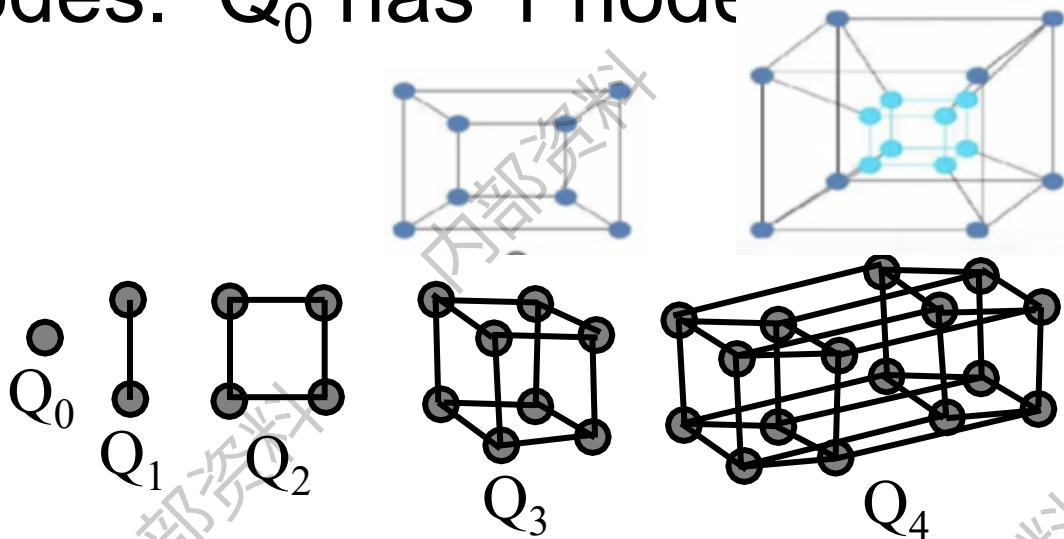
- For any $n \geq 3$, a wheel W_n is a simple graph obtained by taking the cycle C_n and adding one extra vertex v_{hub} and n extra edges $\{\{v_{\text{hub}}, v_1\}, \{v_{\text{hub}}, v_2\}, \dots, \{v_{\text{hub}}, v_n\}\}$.



How many edges are there in W_n ?

n -cubes (hypercubes)

- For any $n \in \mathbf{N}$, the hypercube Q_n is a simple graph consisting of two copies of Q_{n-1} connected together at corresponding nodes. Q_0 has 1 node



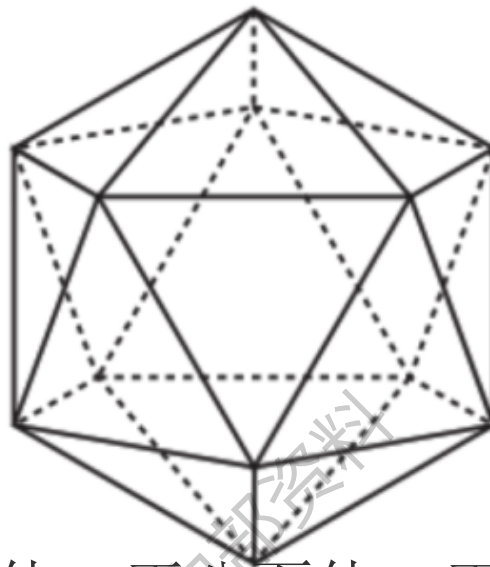
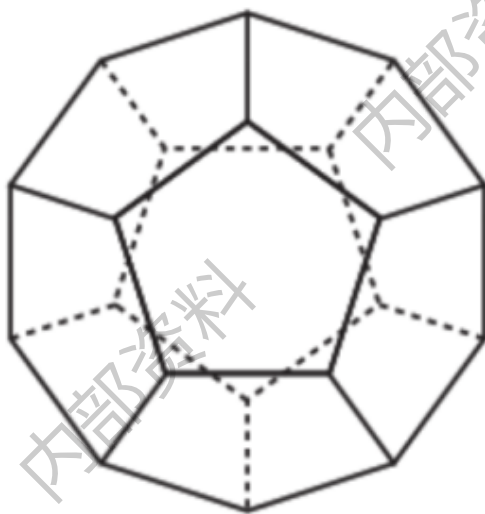
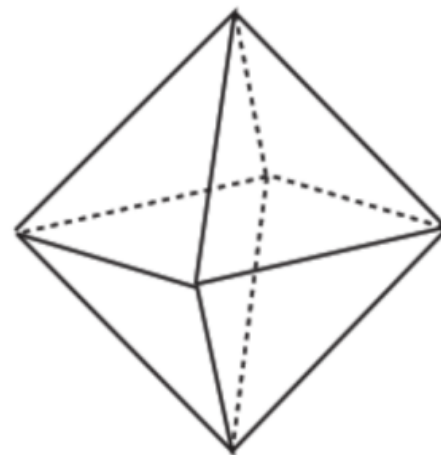
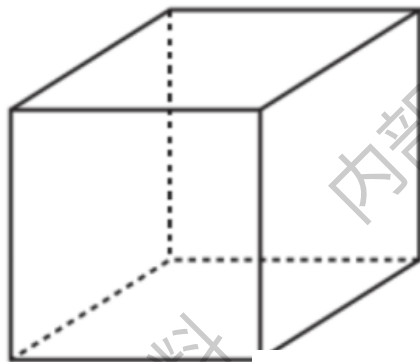
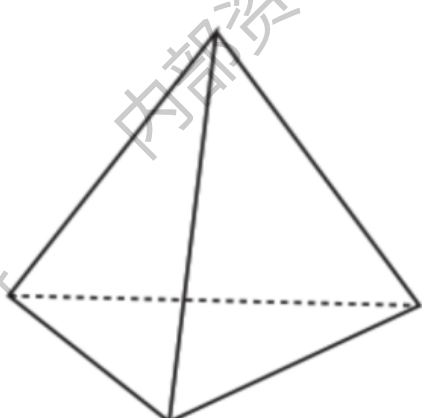
Number of vertices: 2^n . Number of edges: Exercise to try!

$$a_n = 2a_{n-1} + 2^{n-1}$$

n -cubes (hypercubes)

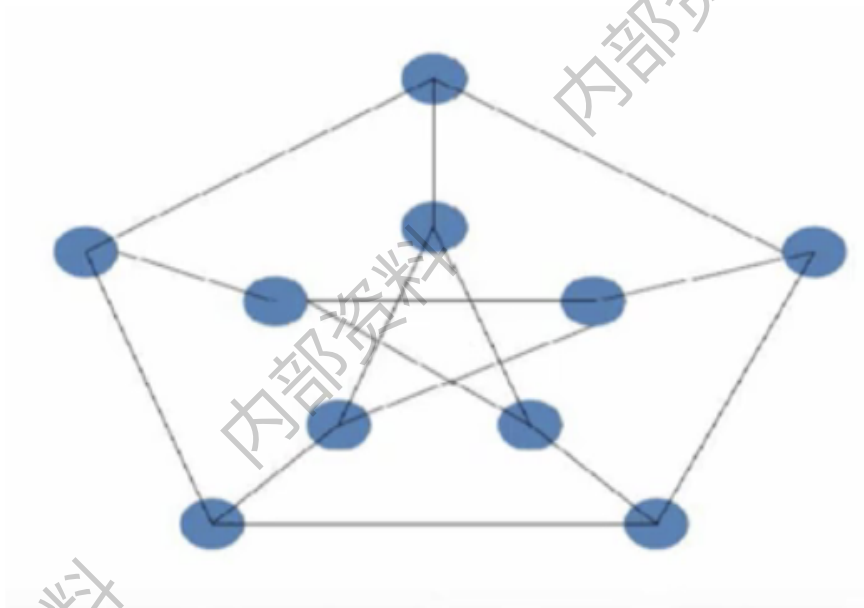
- For any $n \in \mathbf{N}$, the hypercube Q_n can be defined recursively as follows:
 - $Q_0 = \{\{v_0\}, \emptyset\}$ (one node and no edges)
 - For any $n \in \mathbf{N}$, if $Q_n = (V, E)$, where $V = \{v_1, \dots, v_a\}$ and $E = \{e_1, \dots, e_b\}$, then $Q_{n+1} = (V \cup \{v_1', \dots, v_a'\}, E \cup \{e_1', \dots, e_b'\} \cup \{\{v_1, v_1'\}, \{v_2, v_2'\}, \dots, \{v_a, v_a'\}\})$ where $v_1', \dots, v_a'$ are new vertices, and where if $e_i = \{v_j, v_k\}$ then $e_i' = \{v_j', v_k'\}$.

Plato graphs (柏拉图图)



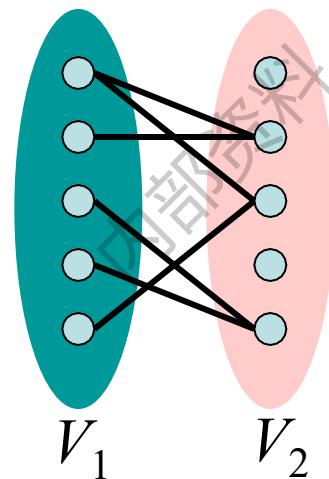
柏拉图图：正四面体、正六面体、正八面体、正十二面体和正二十面体所相应的图的统称

彼得森图 (Petersen)



Bipartite Graphs 二分图

- **Def'n.:** A graph $G=(V,E)$ is *bipartite* (two-part) iff $V = V_1 \cup V_2$ where $V_1 \cap V_2 = \emptyset$ and $\forall e \in E: \exists v_1 \in V_1, v_2 \in V_2: e = \{v_1, v_2\}$.
- The graph can be divided into two parts in such a way that all edges go between the two parts.



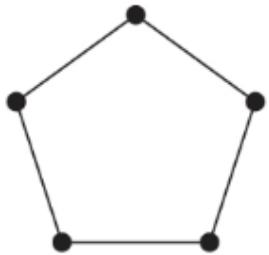
This definition can easily be adapted for the case of multigraphs and directed graphs as well.

Can represent with zero-one matrices.

Example 10

Example: Show that C_6 is bipartite.

Solution: We can partition the vertex set into $V_1 = \{v_1, v_3, v_5\}$ and $V_2 = \{v_2, v_4, v_6\}$ so that every edge of C_6 connects a vertex in V_1 and V_2 .



C_5

bipartite?

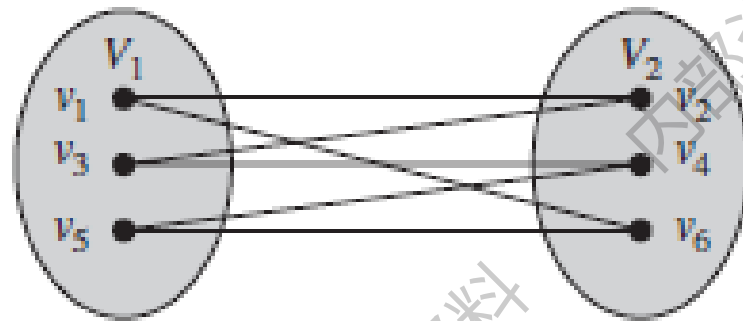


FIGURE 7 Showing That C_6 Is Bipartite.

Theorem

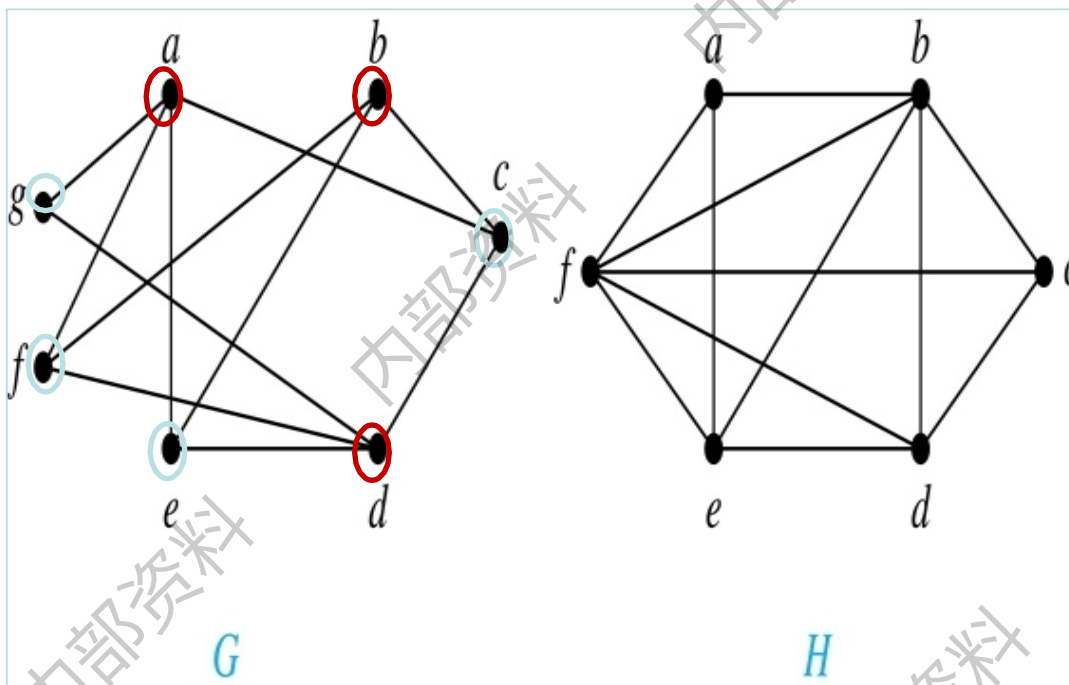
判断准则

- A simple graph is bipartite if and only if it is possible to assign one of two different colors to each vertex of the graph so that no two adjacent vertices are assigned the same color.

补充：充要条件：图中不存在长度为奇数的回路。

Example 11

G is bipartite

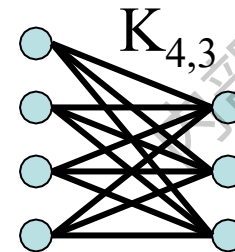


H is not bipartite since if we color a red, then the adjacent vertices f and b must both be blue.

Complete Bipartite Graphs

完全二分图

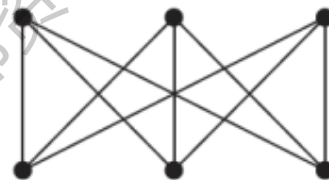
- For $m, n \in \mathbf{N}$, the *complete bipartite graph* $K_{m,n}$ is a bipartite graph where
- $|V_1| = m$,
 $|V_2| = n$, and $E = \{\{v_1, v_2\} | v_1 \in V_1 \wedge v_2 \in V_2\}$.
 - That is, there are m nodes in the left part, n nodes in the right part, and every node in the left part is connected to every node in the right part.



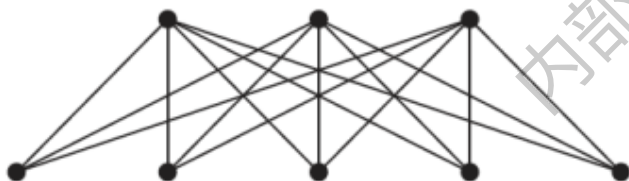
$K_{m,n}$ has _____ nodes
and _____ edges.



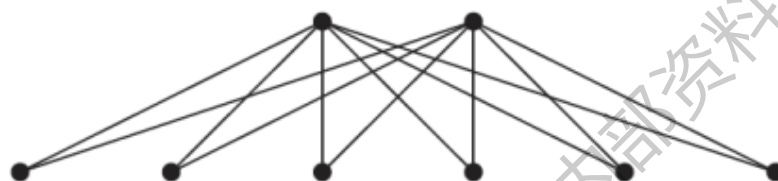
$K_{2,3}$



$K_{3,3}$



$K_{3,5}$



$K_{2,6}$

FIGURE 9 Some Complete Bipartite Graphs.

Bipartite Graphs and Matchings

- Job Assignments

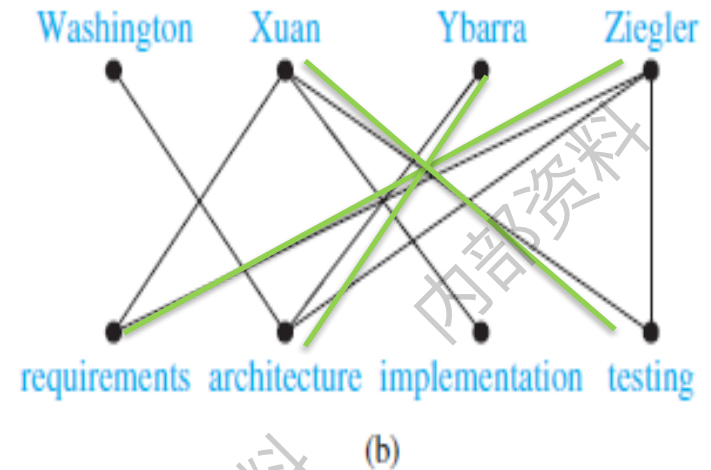
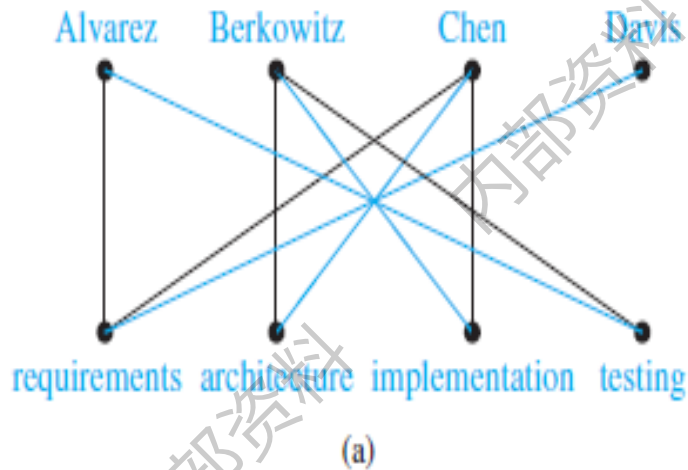


FIGURE 10 Modeling the Jobs for Which Employees Have Been Trained.

Matching

匹配

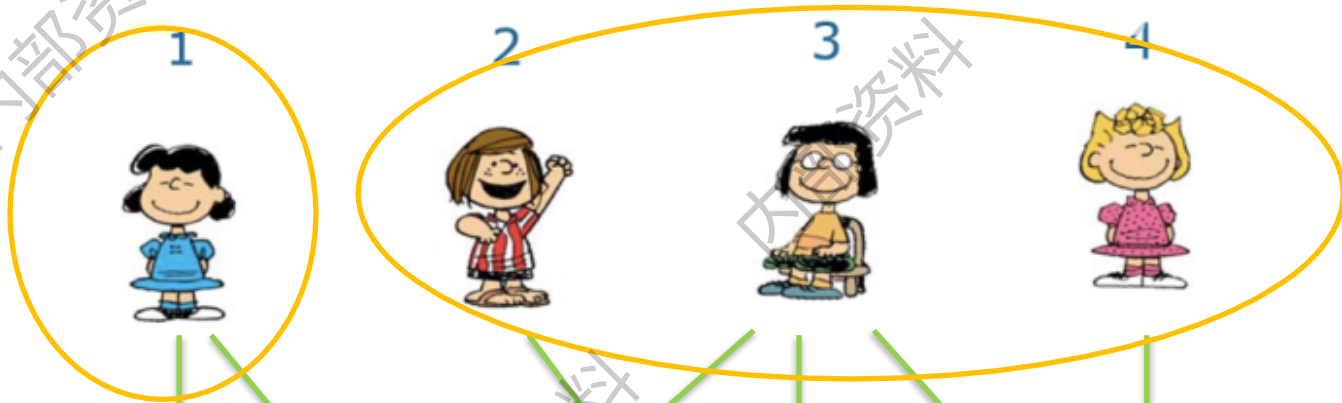
- A matching M in a simple graph $G = (V, E)$ is a **subset** of the set E of edges of the graph such that no two edges are incident with the same vertex.
- A vertex that is the endpoint of an edge of a matching M is said to be **matched in M** ; otherwise it is said to be **unmatched**.
- A **maximum matching** is a matching with the largest number of edges. 最大匹配总是存在的
- A matching M in a bipartite graph $G = (V, E)$ with bipartition (V_1, V_2) is a **complete matching from V_1 to V_2** if every vertex in V_1 is the endpoint of an edge in the matching, or equivalently, if $|M| = |V_1|$. 完全匹配

HALL'S MARRIAGE THEOREM

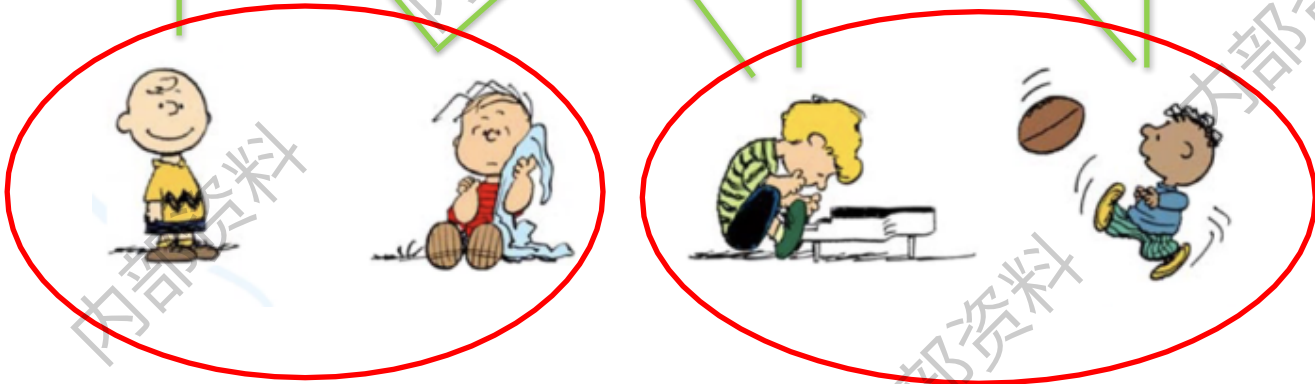
霍尔婚姻定理

- The bipartite graph $G = (V, E)$ with bipartition (V_1, V_2) has a complete matching from V_1 to V_2 if and only if $|N(A)| \geq |A|$ for all subsets A of V_1 .

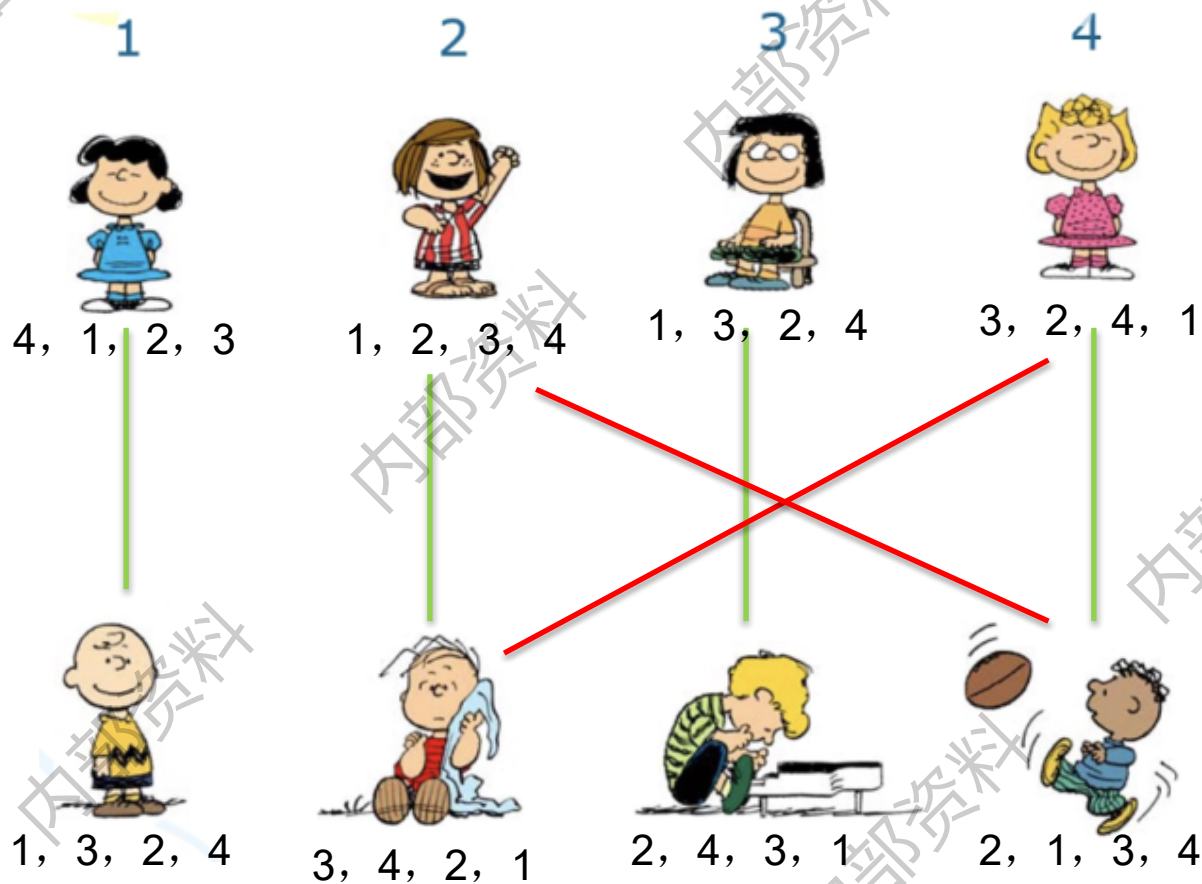
V2



V1



稳定匹配?



延迟决定法



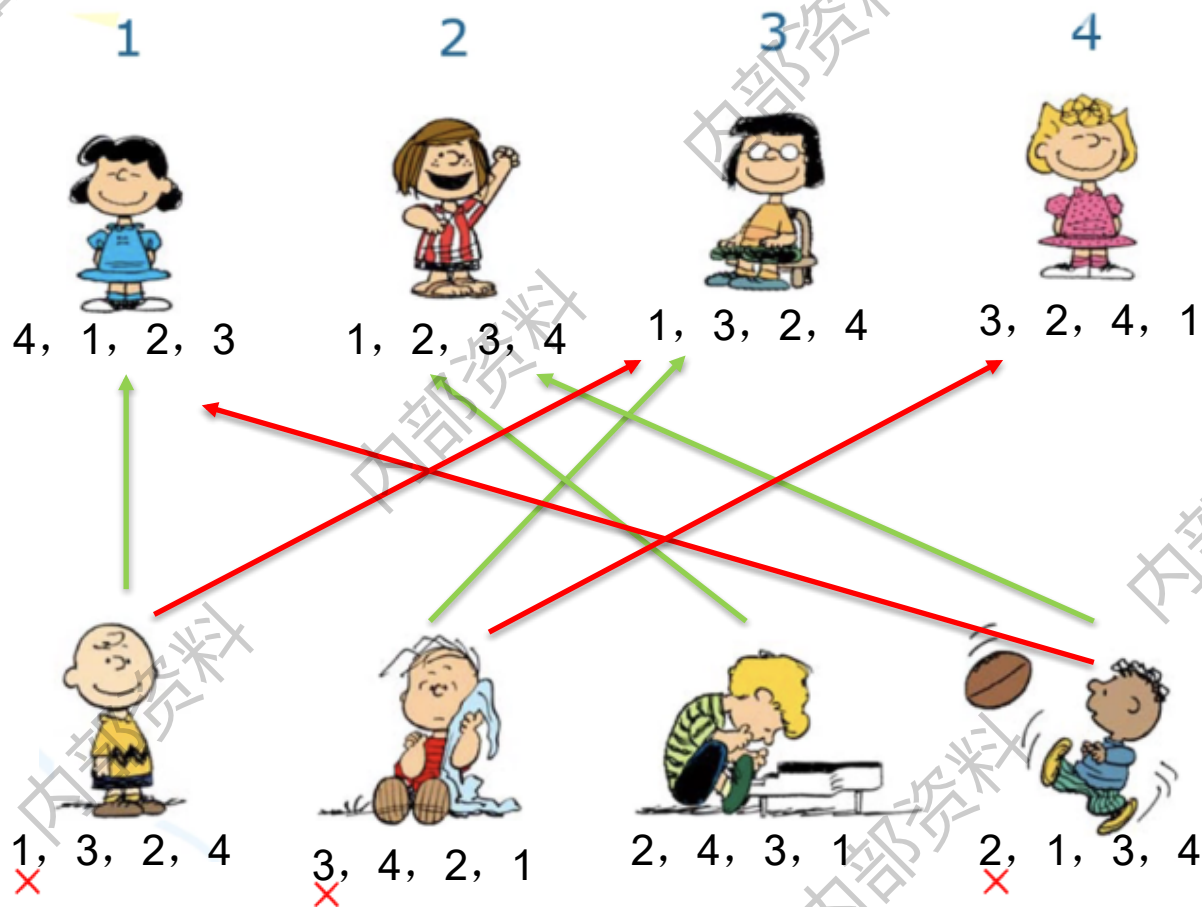
David Gale
大卫·戈尔



Lloyd Shapley
劳埃德·沙普利

- 第一轮：每位男士向他喜欢的女士求婚，女士留下一位求婚者，拒绝其他人
- 第二轮：每位男士向他喜欢的，且未拒绝他的女士求婚。每位女士留下一位喜欢的男士，拒绝其他求婚者
-
- 没有任何男士被拒绝。结束。

稳定匹配



可以证明，如果 n 位男士， n 位女士，最多 n^2 轮可以结束

Some Applications of Special Types of Graphs

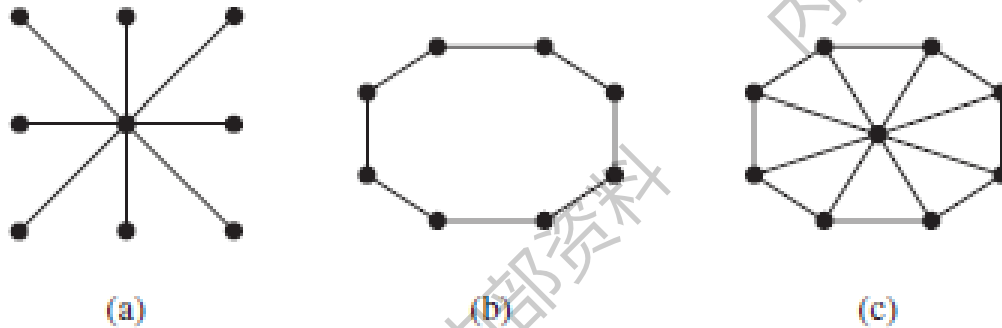


FIGURE 11 Star, Ring, and Hybrid Topologies for Local Area Networks.

Interconnection Networks for Parallel Computation



FIGURE 12 A Linear Array for Six Processors.

Mesh network($n=m^2, p(i,j)$ link $p(i\pm 1,j)$ and $p(i,j\pm 1)$)

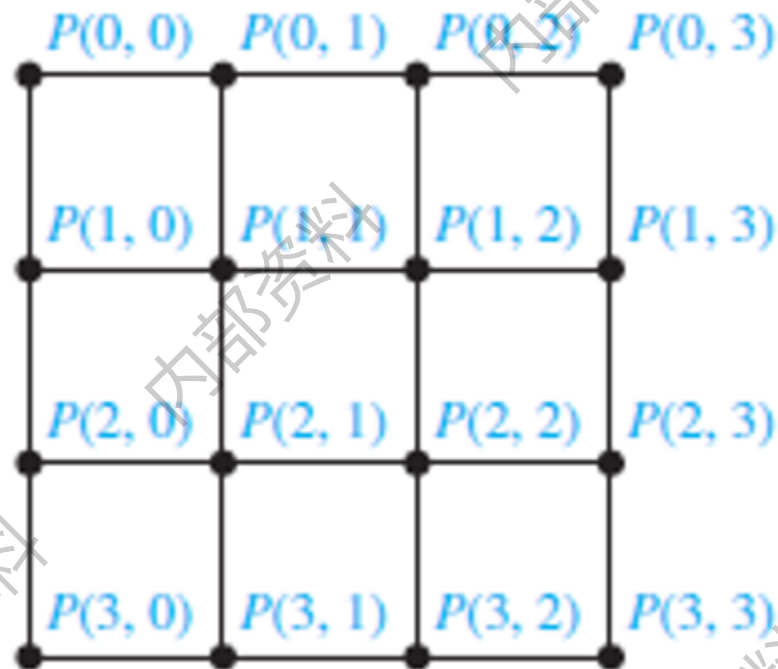


FIGURE 13 A Mesh Network for 16 Processors.

Hypercube $n=2^m$

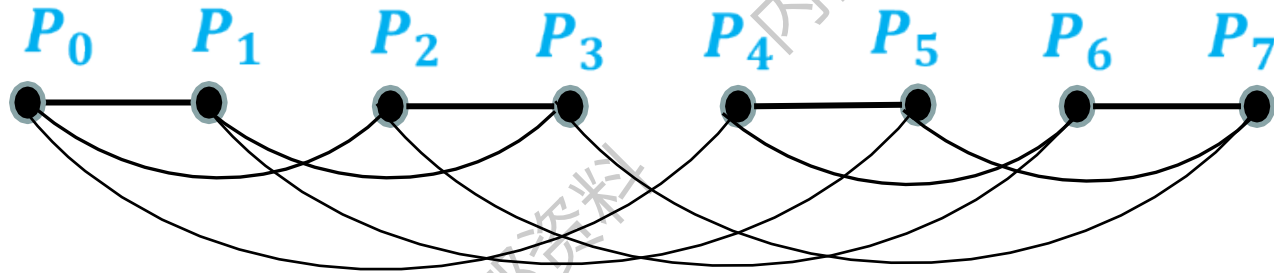
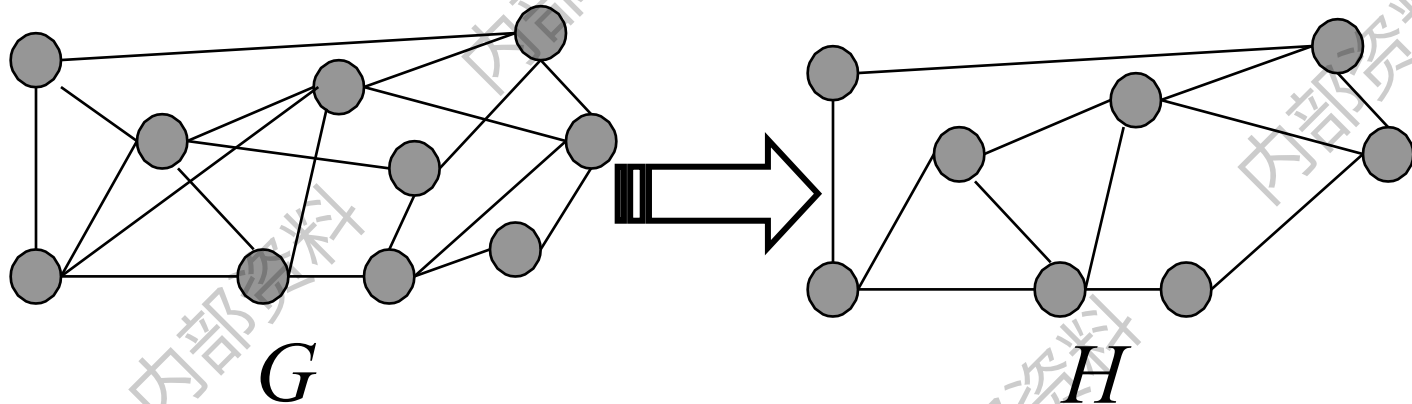


FIGURE 14 A Hypercube Network for Eight Processors.

Subgraphs 子图

- A subgraph of a graph $G=(V,E)$ is a graph $H=(W,F)$ where $W \subseteq V$ and $F \subseteq E$.

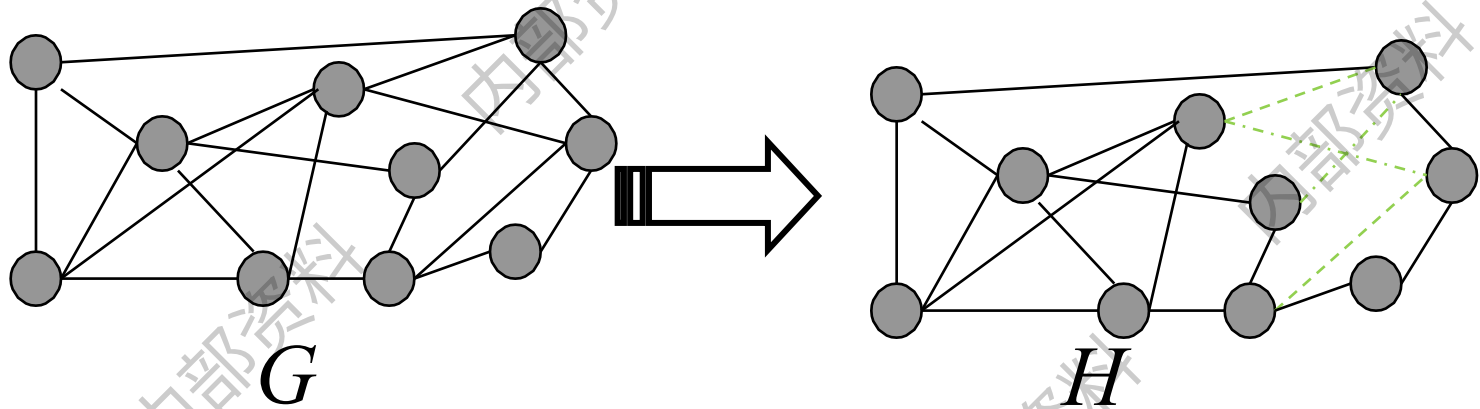


$H \neq G$

H is proper subgraph of G (真子图)

Spanning subgraph 生成子图

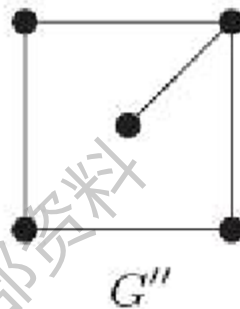
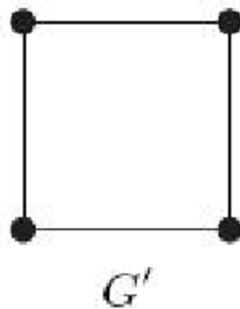
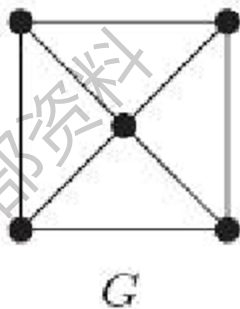
- Deleting edges and not deleting vertices of graph G



Induced Subgraph

导出子图

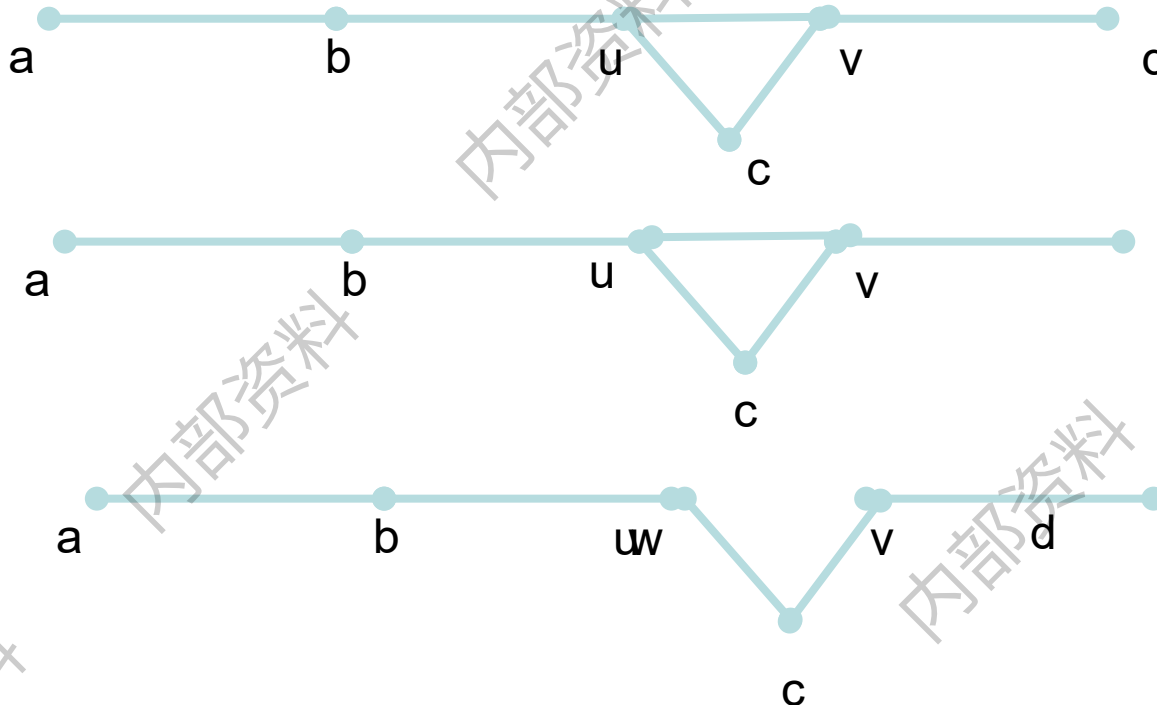
- If $G' \subseteq G$ and G' contains all edges $xy \in E$ with $x, y \in V'$, then G' is an induced subgraph of G . we say that V' **induces or spans G'** in G , and write $G' := G[V']$.



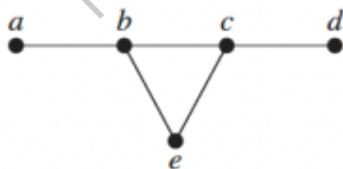
The contraction of G

图的收缩

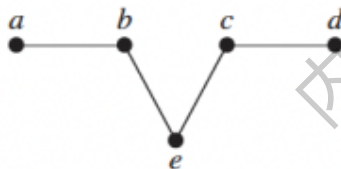
- constructed from G by replacing the edge $\{u, v\}$ with a new vertex w , and replacing the edges $\{u, b\}$, $\{v, d\}$, $\{u, c\}$, and $\{v, c\}$ with the new edges $\{w, b\}$, $\{w, d\}$, and $\{w, c\}$



Example 19

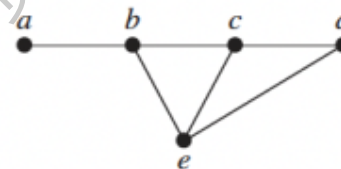


G



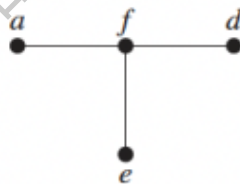
$G - \{b, c\}$

(a)



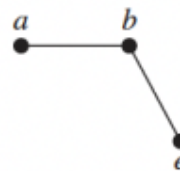
$G + \{e, d\}$

(b)



G contracted by replacing $\{b, c\}$ by f

(c)



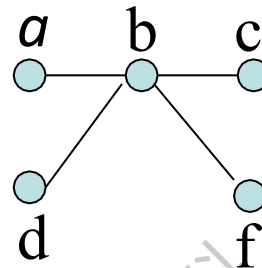
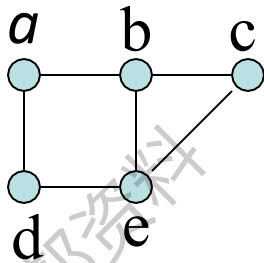
$G - c$

(d)

FIGURE 16 The graph G and four graphs resulting from different operations on G .

Graph Unions

- The *union* $G_1 \cup G_2$ of two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is the simple graph $(V_1 \cup V_2, E_1 \cup E_2)$.

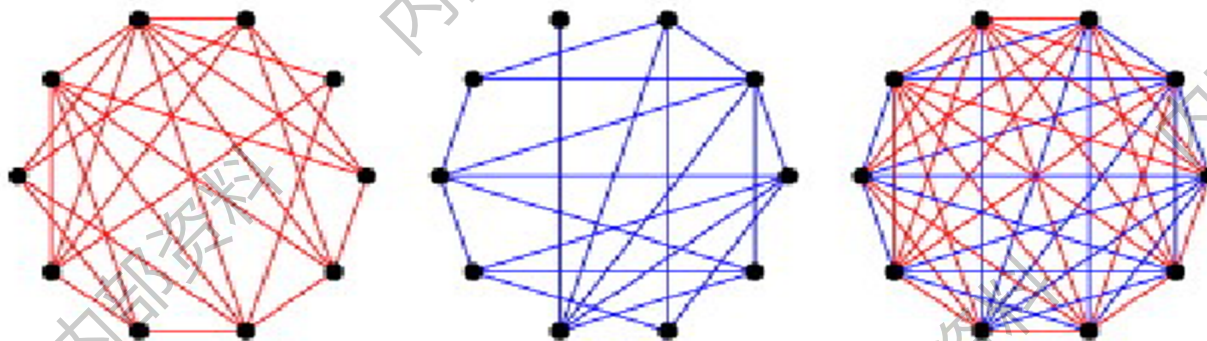


Graph Intersection

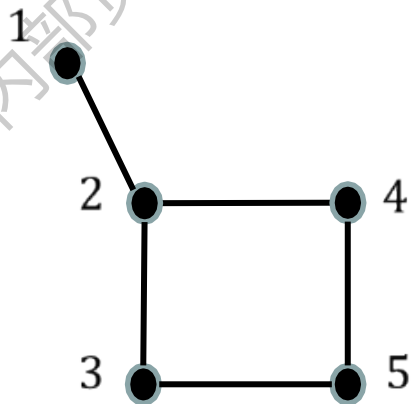
- The intersection $G_1 \cap G_2$ of two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is the simple graph $(V_1 \cap V_2, E_1 \cap E_2)$.
- The difference $G_1 - G_2$ of two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is the simple graph $(V_1 - V_2, E_1 - E_2)$.

Graph complement

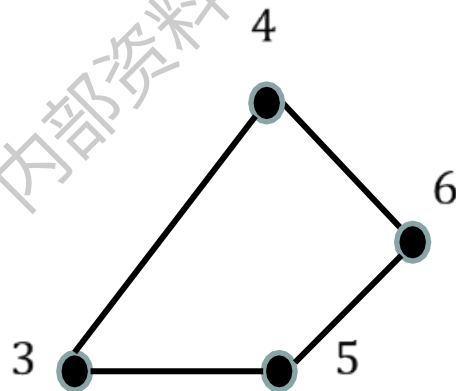
- The complement of a graph G is the graph G' (sometimes denoted $\bar{G}$) with the same vertex set but whose edge set consists of the edges not present in G .



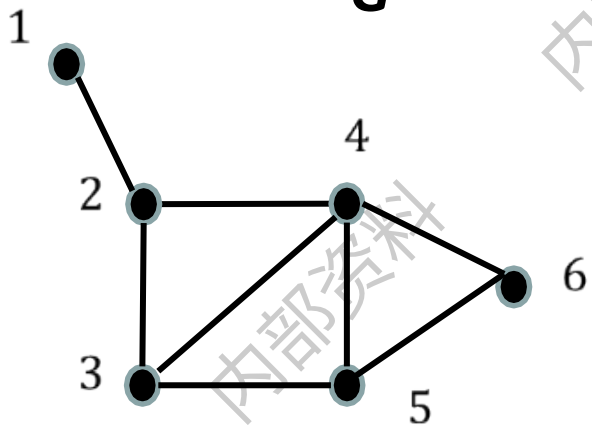
Union, Intersection, difference



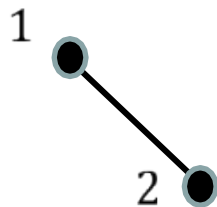
G



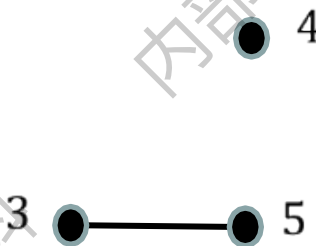
G'



$G \cup G'$



$G - G'$



$G \cap G'$

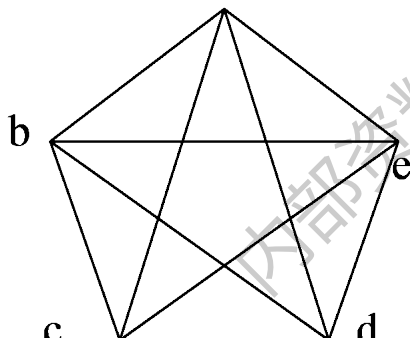


图 A

图 A 是一个无向完全图

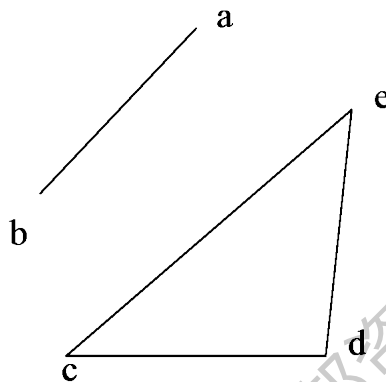


图 B

图 B 的边 $[a, b]$ 是孤立边

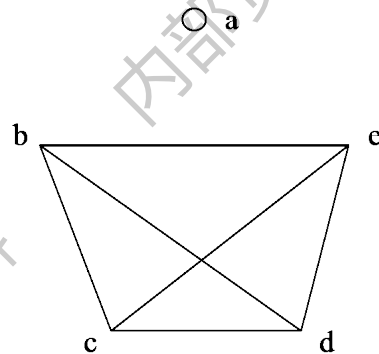


图 C

图 C 的顶点 a 是孤立顶点

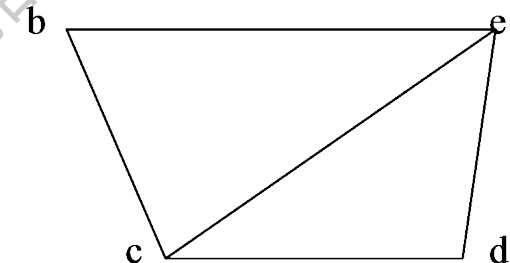


图 D

图 D 是图 C 的子图

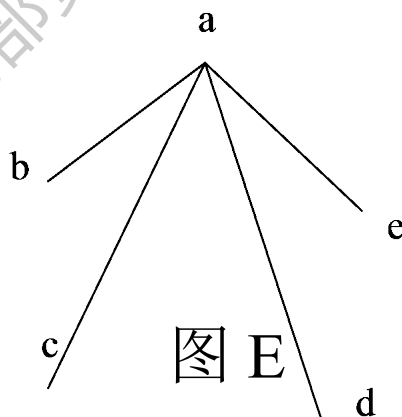


图 E

图 E 是图 C 的补图

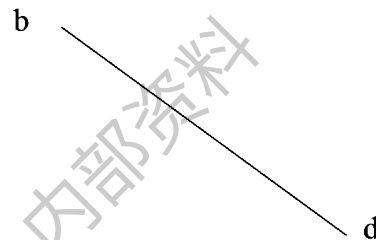


图 F

图 B, C, D, E, F 均是 A 的子图。C、E 为生成子图

Homework

- §10.1 24,36
- §10.2 12, 42, 66-70, 74

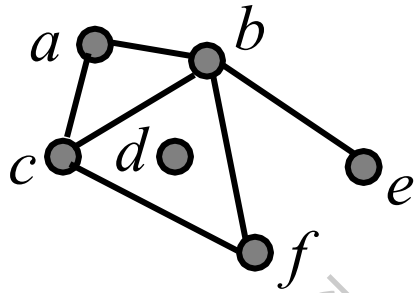
§10.3: Graph Representations & Isomorphism

- Graph representations (图的表示)
 - Adjacency lists. (邻接表)
 - Adjacency matrices. (邻接矩阵)
 - Incidence matrices. (关联矩阵)
- Graph isomorphism: (图同构)
 - Two graphs are isomorphic iff they are identical except for their node names.

Adjacency Lists

邻接表

- A table with 1 row per vertex, listing its adjacent vertices.



<i>Vertex</i>	<i>Adjacent Vertices</i>
<i>a</i>	<i>b, c</i>
<i>b</i>	<i>a, c, e, f</i>
<i>c</i>	<i>a, b, f</i>
<i>d</i>	
<i>e</i>	<i>b</i>
<i>f</i>	<i>c, b</i>

Directed Adjacency Lists

有向邻接表

- 1 row per node, listing the terminal nodes of each edge incident from that node.

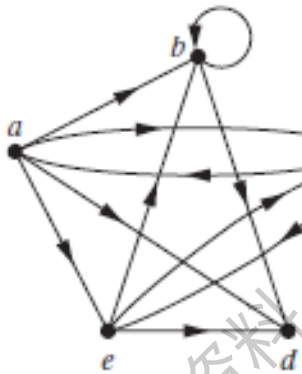


FIGURE 2 A Directed Graph.

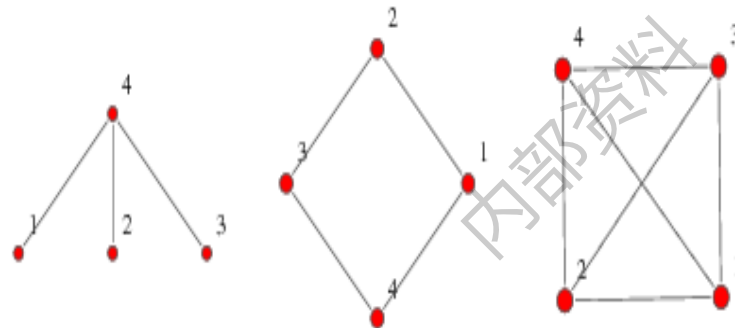
TABLE 2 An Adjacency List for a Directed Graph.

<i>Initial Vertex</i>	<i>Terminal Vertices</i>
<i>a</i>	<i>b, c, d, e</i>
<i>b</i>	<i>b, d</i>
<i>c</i>	<i>a, c, e</i>
<i>d</i>	
<i>e</i>	<i>b, c, d</i>

Adjacency Matrices

邻接矩阵

- A way to represent simple graphs
– possibly with self-loops.
- Matrix $\mathbf{A}=[a_{ij}]$,
$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G, \\ 0 & \text{otherwise.} \end{cases}$$
- Can extend to pseudographs by letting each matrix element be the number of links (possibly >1) between the nodes.



$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

Note

- An adjacency matrix of a graph is based on the ordering chosen for the vertices.
- The adjacency matrix of a simple graph is symmetric.
- $a_{ii}=0$
- Sparse matrix(稀疏矩阵)

Example 4

- Use an adjacency matrix to represent the graph shown in Figure 3.

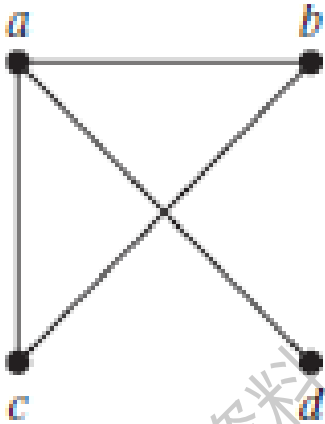
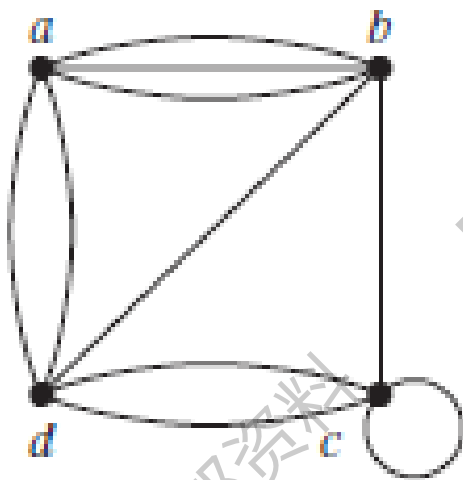


FIGURE 3
Simple Graph.

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Example 5

- Use an adjacency matrix to represent the pseudograph shown in Figure 5.



$$\begin{bmatrix} 0 & 3 & 0 & 2 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix}$$

FIGURE 5

A Pseudograph.

有向图邻接矩阵

- Adjacency matrix for a directed graph

$$a_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \text{ is an edge of } G, \\ 0 & \text{otherwise.} \end{cases}$$

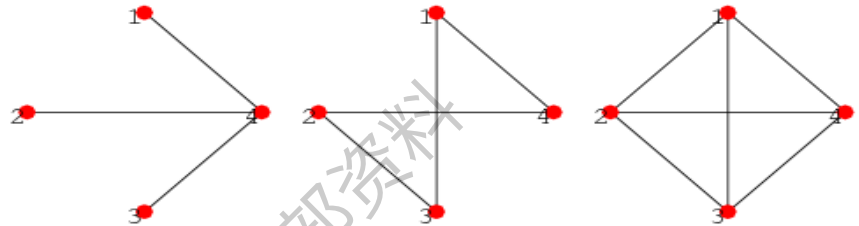
- Adjacency matrix for a directed multigraph

Incidence matrices

关联矩阵

- Let $G(V, E)$ be an **undirected graph**.
- Suppose that $v_1, v_2, \dots, v_n$ are the vertices and $e_1, e_2, \dots, e_m$ are edges of G .
- Then the incidence matrix with respect to this ordering of V and E is the $n \times m$ matrix $M = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i, \\ 0 & \text{otherwise} \end{cases}$$



$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$$

Example 6

- Represent the graph shown in Figure 6 with an incidence matrix.

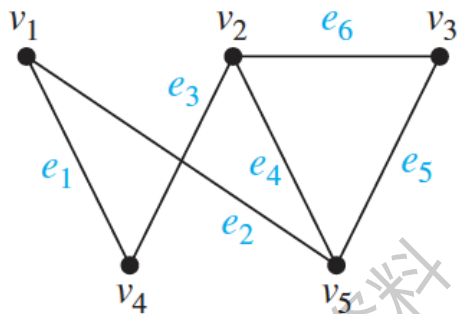


FIGURE 6 An Undirected Graph.

$$\begin{array}{c}
 e_1 \quad e_2 \quad e_3 \quad e_4 \quad e_5 \quad e_6 \\
 \begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array}
 \begin{bmatrix}
 1 & 1 & 0 & 0 & 0 & 0 \\
 0 & 0 & 1 & 1 & 0 & 1 \\
 0 & 0 & 0 & 0 & 1 & 1 \\
 1 & 0 & 1 & 0 & 0 & 0 \\
 0 & 1 & 0 & 1 & 1 & 0
 \end{bmatrix}
 \end{array}$$

Example 7

- Represent the pseudograph shown in Figure 7 using an incidence matrix.

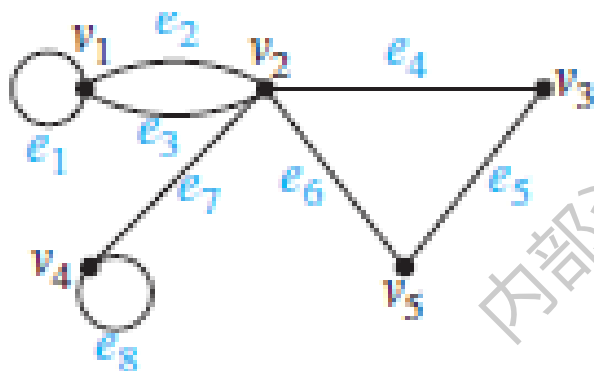


FIGURE 7
A Pseudograph.

	e_1	e_2	e_3	e_4	e_5	e_6	e_7	e_8
v_1	1	1	1	0	0	0	0	0
v_2	0	1	1	1	0	1	1	0
v_3	0	0	0	1	1	0	0	0
v_4	0	0	0	0	0	0	1	1
v_5	0	0	0	0	1	1	0	0

Graph Isomorphism

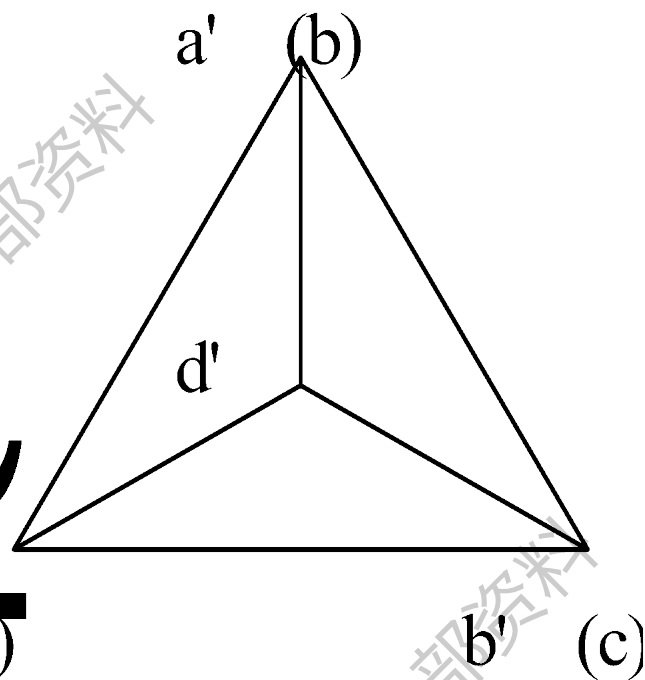
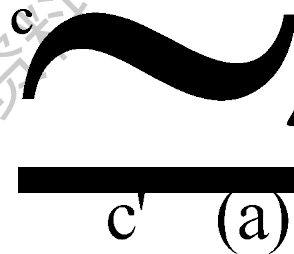
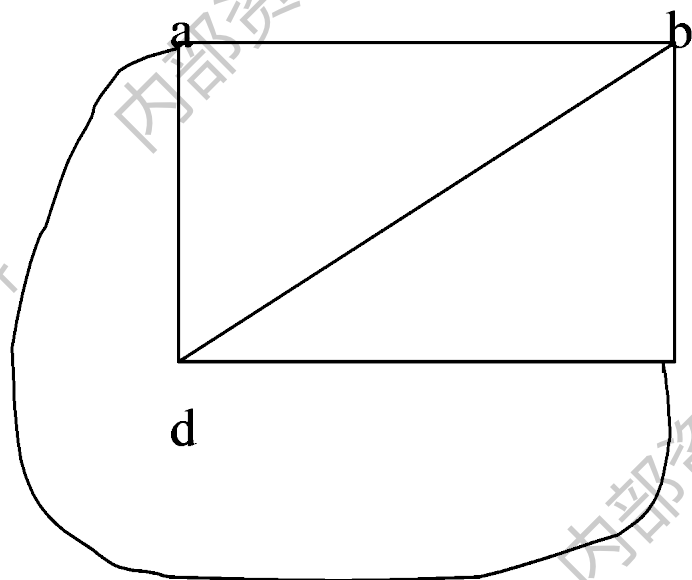
图同构

- Formal definition:
 - Simple graphs $G_1=(V_1, E_1)$ and $G_2=(V_2, E_2)$ are *isomorphic* iff $\exists$ a bijection $f:V_1 \rightarrow V_2$ such that $\forall a,b \in V_1$, a and b are adjacent in G_1 iff $f(a)$ and $f(b)$ are adjacent in G_2 .
 - f is the “renaming” function between the two node sets that makes the two graphs identical.
 - This definition can easily be extended to other types of graphs.

Nauty--One of the algorithms for judging isomorphism

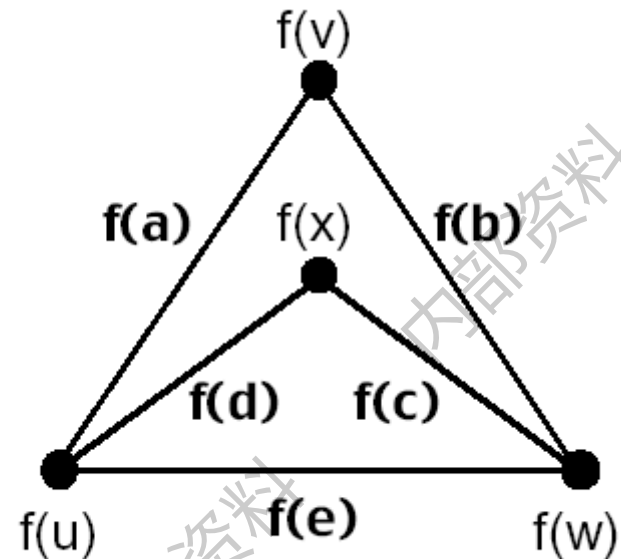
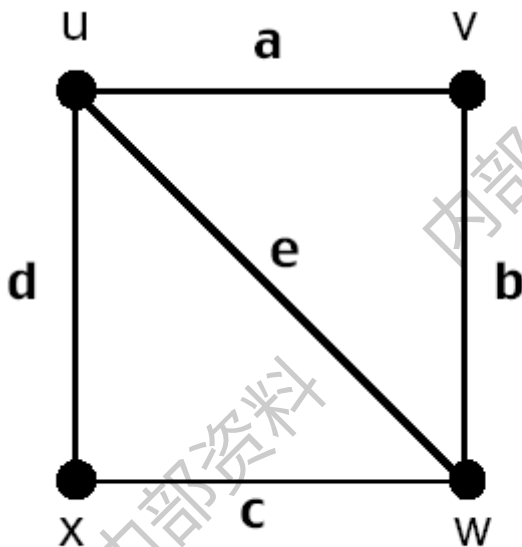
example

无向图

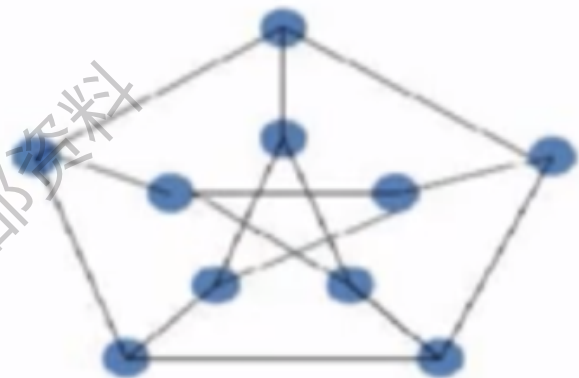


Example

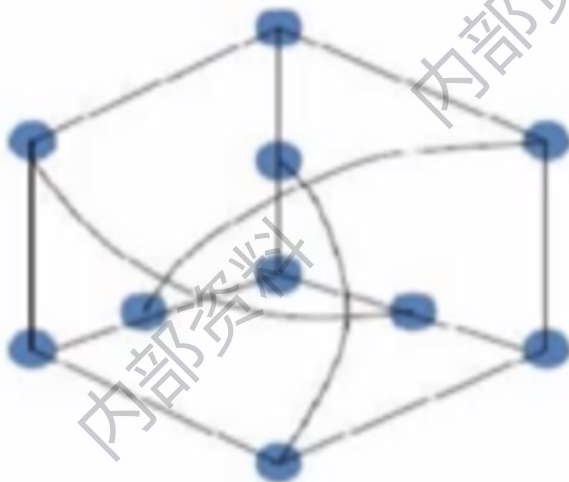
- The graph mapping f is an isomorphism.



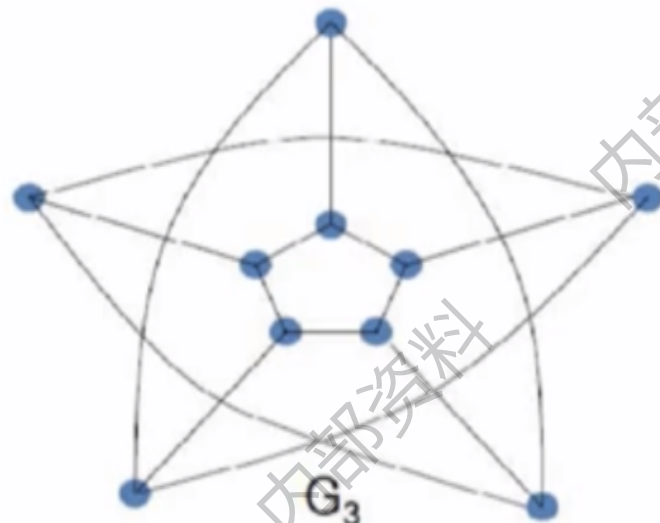
Petersen



G_1



G_2



G_3

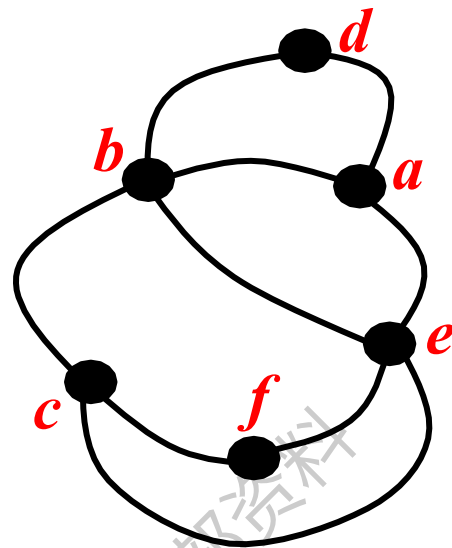
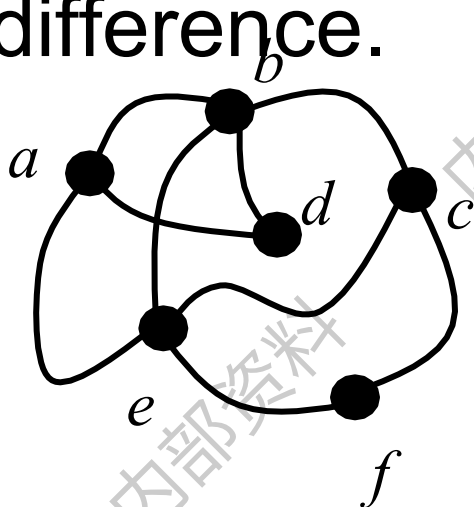
Graph Invariants (图形不变量) under Isomorphism

Necessary but not sufficient conditions for $G_1=(V_1, E_1)$ to be isomorphic to $G_2=(V_2, E_2)$:

- We must have that $|V_1|=|V_2|$, and $|E_1|=|E_2|$.
- The number of vertices with degree n is the same in both graphs.
- For **every proper subgraph** g of one graph, there is a proper subgraph of the other graph that is isomorphic to g .

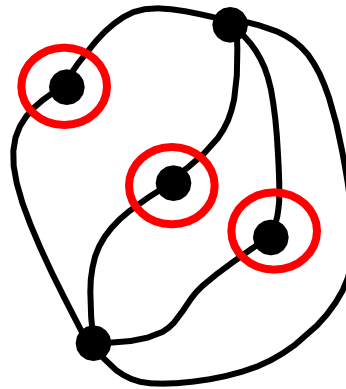
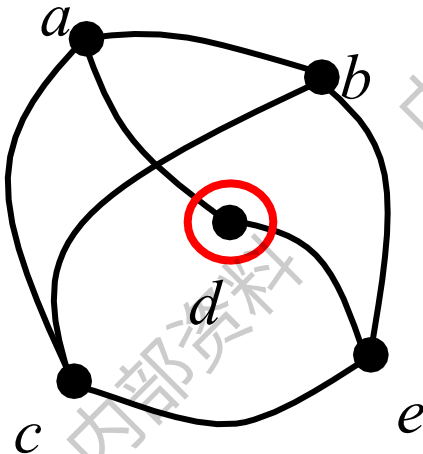
Isomorphism Example

- If isomorphic, label the 2nd graph to show the isomorphism, else identify difference.

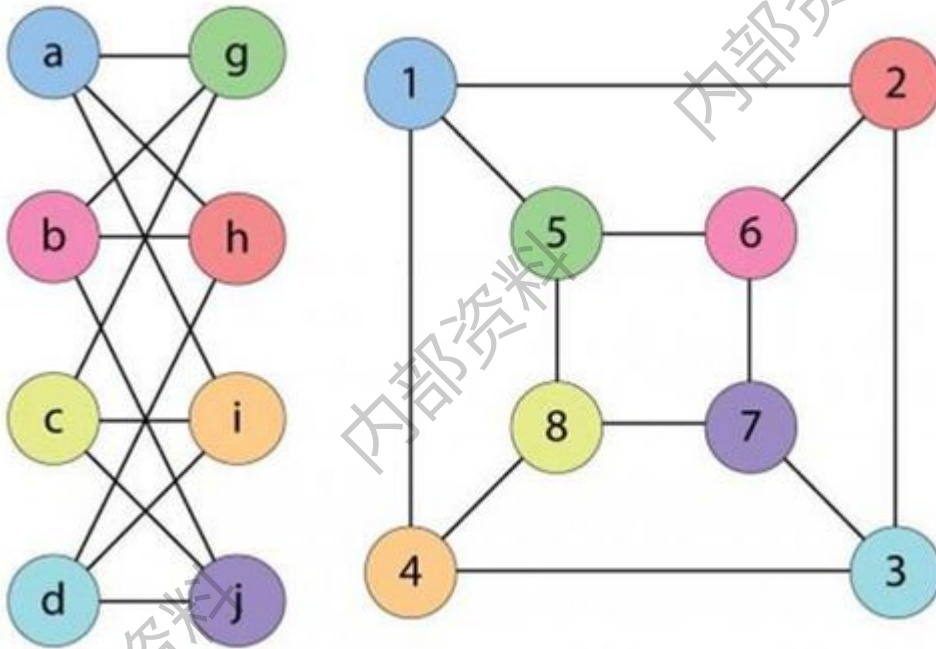


Are These Isomorphic?

- If isomorphic, label the 2nd graph to show the isomorphism, else identify difference.



- *Same # of vertices*
- *Same # of edges*
- *Different # of verts of degree 2! (1 vs 3)*



https://www.sohu.com/a/299438284_120054637?qq-pf-to=pcqq.group

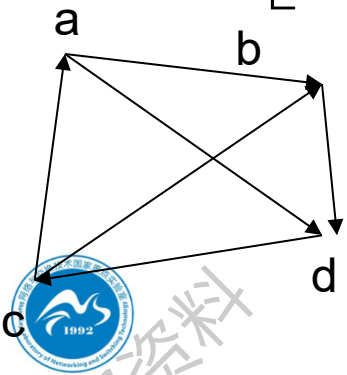
从图中不容易直接得出。

可用邻接矩阵，通过某种算法确定两个图是否同构。

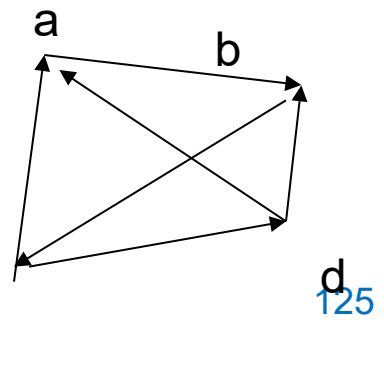
下图的邻接矩阵：

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$



i



同构判定算法（用邻接矩阵）

- 1、根据图确定其邻接矩阵（ I ）
- 2、计算行度：矩阵每行 1 的个数，（对应于出度）
和列度：（对应于入度）
- 3、不考虑出现的次序不同，若行度与列度不同，
则必不同构，否则继续
- 4、同时交换其一矩阵的 i 行和 j 行， i 列和 j 列。

若此矩阵能变成与另一矩阵一样，则同构。对所有顶点的排列都试过，仍不相同，则不同构。

例：(1) 的行度 (2 , 1 , 2 , 1)

列度 (1 , 2 , 1 , 2)

(2) 的行度 (1 , 1 , 2 , 2)

列度 (2 , 2 , 1 , 1)

例： 把(II) 的第 1 行与第 4 行交换，
第 1 列与第 4 列交换，得：

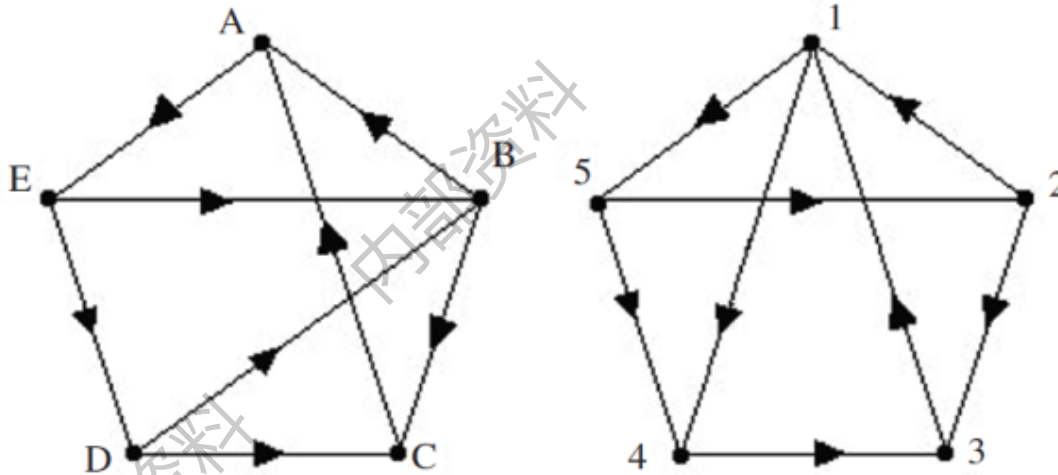
$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

再把第 2 行与第 4 行交换，第 2 列与第 4 列交换，就得：

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Example

Determine whether the following digraphs are isomorphic.



the two digraphs are not isomorphic

Homework

- §10.3
 - 32,50,56,66,74

§10.4: Connectivity

连通性

- Definition: (Paths (通路) in Graphe Theory)
- In an undirected graph $G=(V, E)$, a *path* of length n from $u(=v_0)$ to $v(=v_n)$ is a sequence of adjacent edges $e_1, e_2, \dots, e_n$, going from vertex u to vertex v , such that $e_i=\{v_{i-1}, v_i\}$ for $1 \leq i \leq n$.
- is a sequence of vertex $v_0, v_1, \dots, v_n$ such that $\{v_i, v_{i+1}\} \in E$ for $0 \leq i \leq n-1$.

- undirected graph
 - A path is a **circuit** (回路) if **$u=v$ and $n>0$** .
 - A path pass through the vertices or traverses the edges. 遍历
 - A path is *simple* if it contains no edge more than once.

Example 1

- A simple path a, d, c, f, e
- Not a path: d, e, c, b
 $\{e, c\}$ no edge
- path: a, b, e, d, a, b , **not simple**
- circuit: a, e, f, b, a ,
length 4

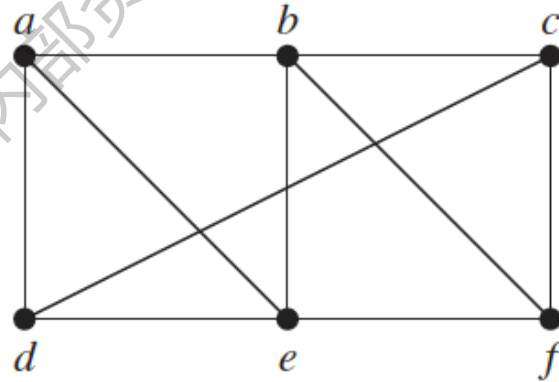


FIGURE 1 A Simple Graph.

Paths in Directed Graphs

有向图的通路

- Same as in undirected graphs, but the path must go in the direction of the arrows.

Connectedness

连通图

- An undirected graph is *connected* if there is a path between every pair of distinct vertices in the graph.

G_1 connected
 G_2 disconnected

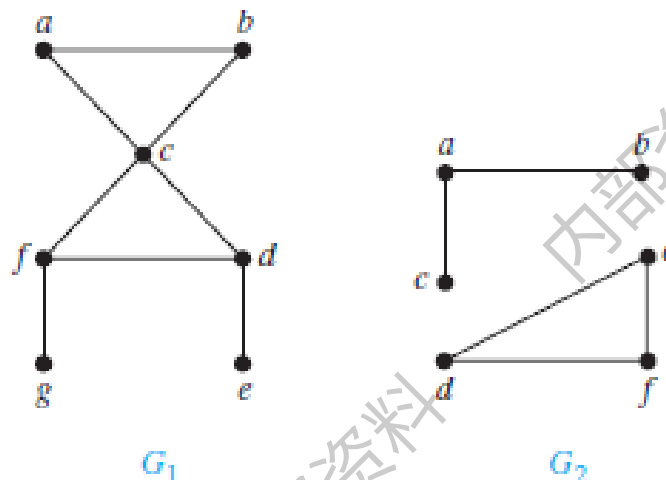


FIGURE 2 The Graphs G_1 and G_2 .

- **Theorem 1:** There is a *simple path* between every pair of distinct vertices of a connected undirected graph.

Connected component 连通分支/独立子图

- A *Connected component* of a graph G is a connected subgraph of G that is not a proper subgraph of another connected subgraph of G .
- A graph G that is not connected has two or more connected components that are disjoint and have G as their union.

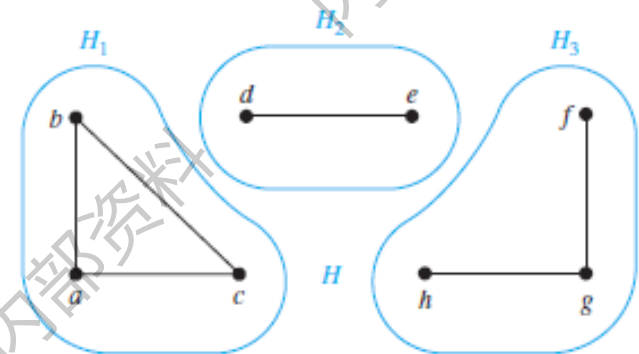
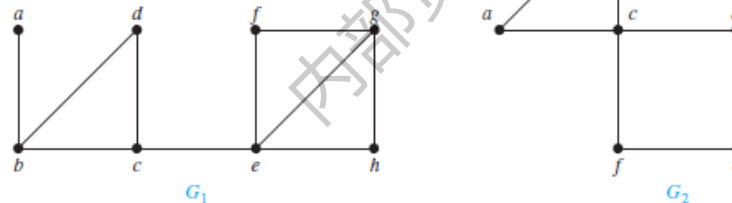


FIGURE 3 The Graph H and Its Connected Components H_1 , H_2 , and H_3 .

Connectivity

(连通度)



- **cut vertex** (割点): the removal from a graph of a vertex and all incident edges produces a subgraph with more connected components.
- **cut edge (bridge)** (割边): an edge whose removal produces a graph with more connected components
- **Vertex cut** (点割集)
- **edge cut** (边割集)

- Vertex Connectivity (点连通度) -- $\kappa(G)$
 - $\kappa(G)$ is minimum number of vertices that can be removed from G to either disconnect G or produce a graph with a single vertex.
 - Nonseparable graphs: without cut vertices.
 - G has n vertices : $0 \leq \kappa(G) \leq n - 1$, $\kappa(G) = 0$ if and only if G is disconnected or $G = K_1$, and $\kappa(G) = n - 1$ if and only if G is complete
 - Vertex cut: $G - V'$ is disconnected.
 - Vertex connectivity: $\kappa(G) = \min |V'|$

- **Edge Connectivity (边连通度) -- $\lambda(G)$**
 - is the minimum number of edges in an edge cut of G
 - Edge cut: $G-E'$ is disconnected.
 - Edge connectivity: $\lambda(G)=\min|E'|$

example8

if $\kappa(G) \geq k$, k -connected

$$\kappa(G_1)=1$$

$$\kappa(G_2)=1$$

$$\kappa(G_3)=2, \text{ 点割集}=\{b,g\}, \{c,f\}$$

$$\kappa(G_4)=2, \text{ 点割集}=\{c,f\}$$

$$\kappa(G_5)=3, \text{ 点割集}=\{b,c,f\}$$

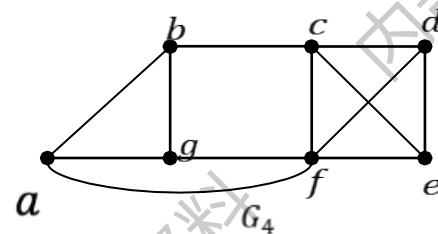
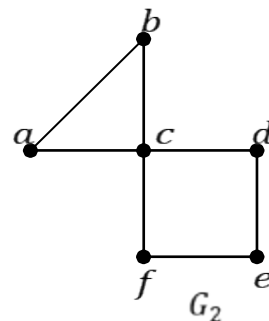
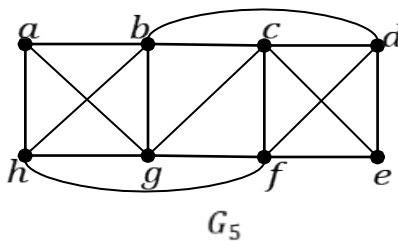
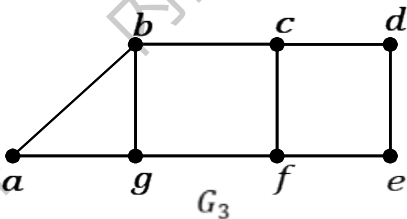
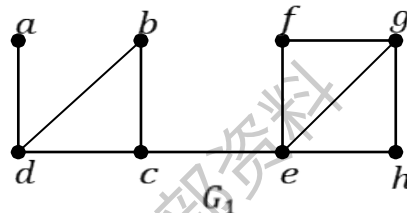
$$\lambda(G_1)=1$$

$$\lambda(G_2)=2, \text{ 边割集}=\{a,b\}, \{a,c\}$$

$$\lambda(G_3)=2, \text{ 边割集}=\{f,g\}, \{b,c\}$$

$$\lambda(G_4)=3, \text{ 边割集}=\{b,c\}, \{a,f\}, \{f,g\}$$

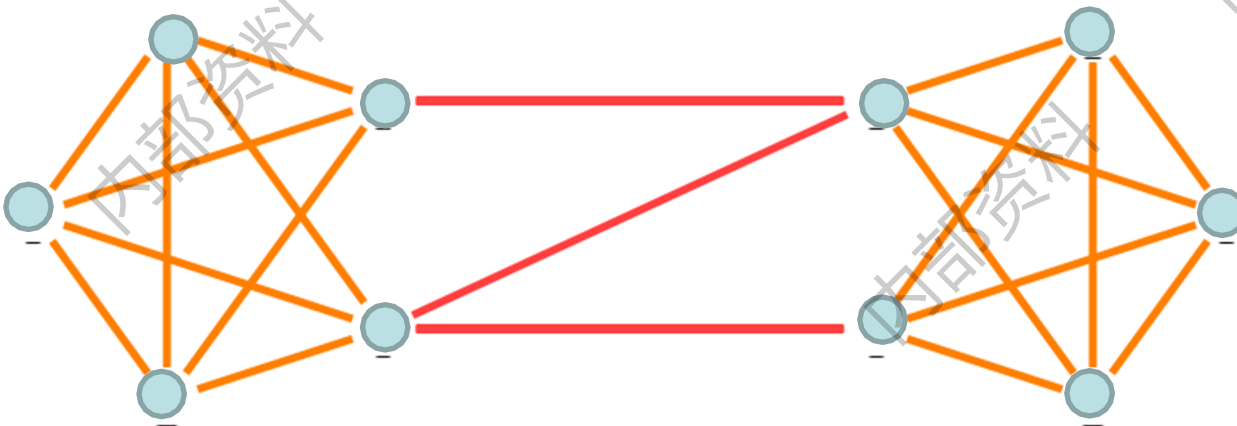
$$\lambda(G_5)=3, \text{ 边割集}=\{a,b\}, \{a,g\}, \{a,h\}$$



Theroem

- Let $\kappa(G)$ be the vertex connectivity of a graph G , $\lambda(G)$ be the edge connectivity, and $\delta(G)$ be the minumum degree, then for every graph,

$$\kappa(G) \leq \lambda(G) \leq \delta(G)$$



Directed Connectedness

- A directed graph is *strongly connected* (强连通) if there is a path from a to b and from b to a whenever a and b are vertices in the graph.
- It is *weakly connected* (弱连通) if there is a path between every two vertices in the underlying undirected graph.
- Note *strongly* implies *weakly* but not vice-versa.

补充定理

- 有向图中，如果存在一个经过所有顶点的连通回路，则为强连通图

Example 10, 11

- Strongly connected components or strong components

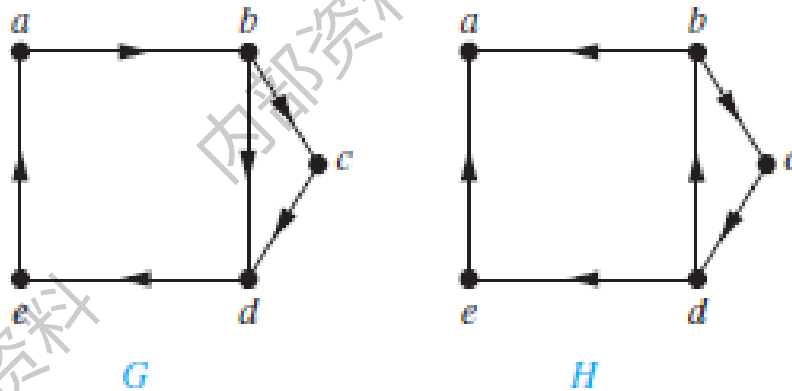


FIGURE 5 The Directed Graphs G and H .

Paths & Isomorphism

- Note that connectedness, and the existence of a circuit or simple circuit of length k are graph invariants with respect to isomorphism.
- Example 13

Invariants: ?

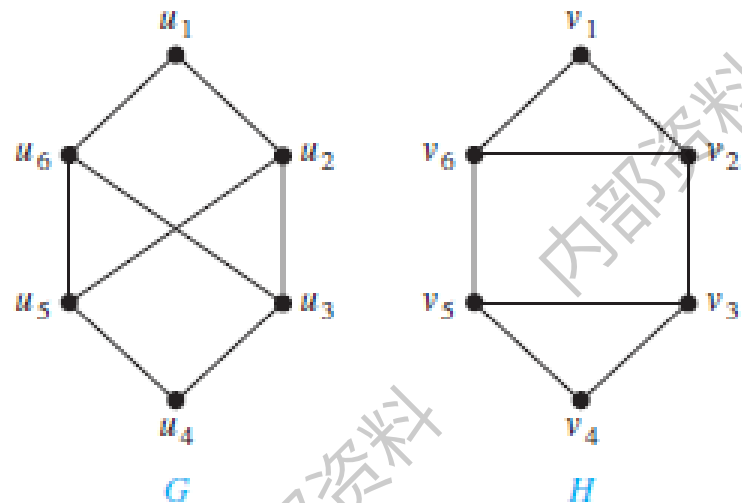


FIGURE 6 The Graphs G and H .

Use paths to find mappings

- Example 14

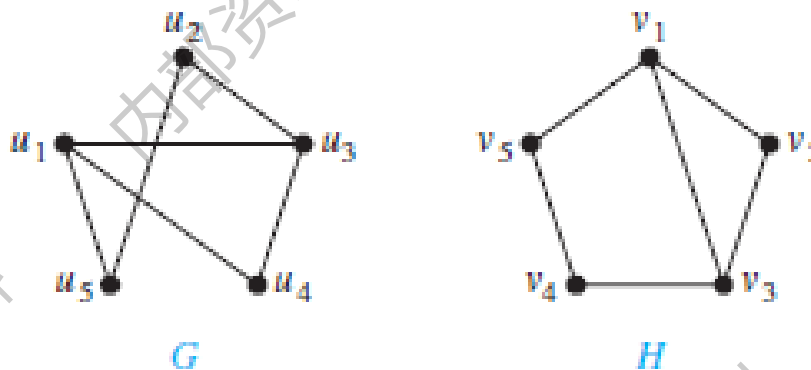
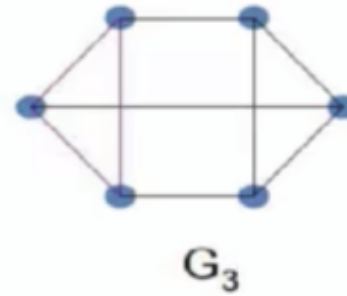
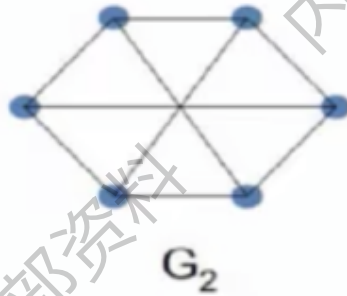
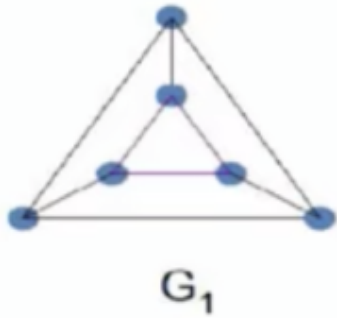


FIGURE 7 The Graphs G and H .

Example



$$G_1 \cong G_3, \quad G_1 \not\cong G_2$$

Counting Paths by Adjacency Matrices

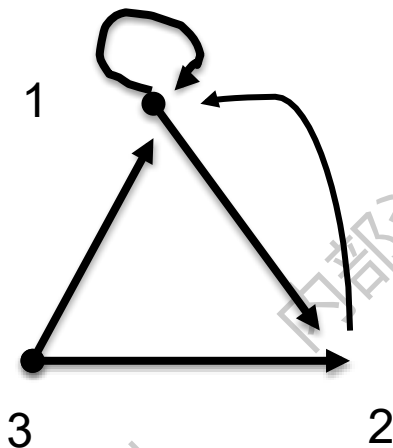
计算两点间通路数

- Let $\mathbf{A}$ be the adjacency matrix of graph G .
- The number of paths of length k from v_i to v_j is equal to $(\mathbf{A}^k)_{i,j}$.
 - The notation $(\mathbf{M})_{i,j}$ denotes $m_{i,j}$ where $[m_{i,j}] = \mathbf{M}$.
 - note: $b_{i1}a_{1j} + b_{i2}a_{2j} + \dots + b_{in}a_{nj}$

$m_{i,j}$ --- 第一个矩阵的 i 行与第二个矩阵的 j 列对应乘积后求和



$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 \times 1 + 1 \times 1 + 0 \times 1 & 1 \times 1 + 1 \times 0 + 0 \times 1 & 1 \times 0 + 1 \times 0 + 0 \times 0 \\ 1 \times 1 + 0 \times 1 + 0 \times 1 & 1 \times 1 + 0 \times 0 + 0 \times 1 & 1 \times 0 + 0 \times 0 + 0 \times 0 \\ 1 \times 1 + 1 \times 1 + 0 \times 1 & 1 \times 1 + 1 \times 0 + 0 \times 1 & 1 \times 0 + 1 \times 0 + 0 \times 0 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 2 & 1 & 0 \end{bmatrix}, \end{aligned}$$



Example 15

- How many paths of length four are there from a to d in the simple graph G in Figure 8?
- a 和 d 之间长度为4的通路数

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

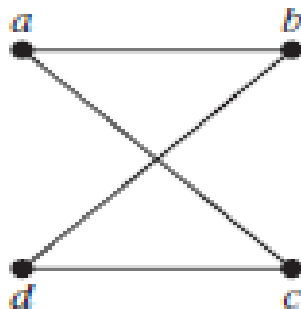


FIGURE 8
Graph G .

$$A^4 = \begin{bmatrix} 8 & 0 & 0 & 8 \\ 0 & 8 & 8 & 0 \\ 0 & 8 & 8 & 0 \\ 8 & 0 & 0 & 8 \end{bmatrix}$$

作业

- §10.1 24,36
- §10.2 12, 44, 66-70, 74
- § 10.3 32,50,56,66,74 (a)
- §10.4 14,28,36,60